

# 生态研究前沿速览\_2026-05-23

## 生态研究前沿速览

2026-05-23

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## 保护与生物多样性

1. Improved forest protection at Chinese heritage relic sites  
(中国遗产遗迹地提升森林保护成效)

Nature Sustainability | 2026-05-22 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

### Abstract

Rapid deforestation and accelerating human pressures throughout the twentieth century reshaped forest landscapes worldwide. Using a national database of 363,282 heritage relic sites in China, historical forest coverage data and a high-resolution human footprint index, the authors show that heritage relics can function as long-term forest refugia. Heritage trees and traditional villages are associated with higher forest coverage and faster regeneration, especially in southern China and lower-pressure regions, but the effect weakens as human footprints intensify.

### 中文摘要

研究整合中国 363,282 个遗产遗迹点、1900-2020 年历史森林覆盖数据和 2020 年高分辨率人类足迹指数，发现遗产树和传统村落等生物文化遗产可作为长期森林避难所，在毁林时期维持森林、在大规模恢复时期促进再生。其保护效应在华南和人类压力较低区域更明显，但会随人类足迹增强而减弱。

### 深度解读

这项研究把文化遗产保护与森林生态韧性直接连接起来，说明传统村落、古树和宗教文化空间并非只是文化资产，也可能是长期维持生态系统服务的基础设施。它为中国及全球生物多样性保护提供了低成本、可嵌入国土空间治理的路径。关键限制在于保护效应依赖周边人类压力，因此遗产保护若不与土地利用管控联动，生态收益会被快速削弱。

DOI: <https://doi.org/10.1038/s41893-026-01846-3>

## 环境与气候

2. A multidimensional assessment of systemic cooling poverty in the global south

(全球南方系统性降温贫困的多维评估)

Nature Sustainability | 2026-05-22 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Extreme heat increasingly threatens human health where climate exposure intersects with social, infrastructural and institutional deprivation. Combining household survey data from 1,155,106 households, climate records and infrastructure datasets, the study builds a systemic cooling poverty index across 28 mostly global-south countries. The analysis finds that almost 600 million people experience high deprivation across cooling-related dimensions.

#### 中文摘要

研究结合 28 个以全球南方为主国家的 1,155,106 户家庭调查、气候记录和基础设施数据，构建系统性降温贫困指数，覆盖气候暴露、基础设施与资产、社会和不平等、健康、教育与工作标准五个维度。结果显示，近 6 亿人处于较高降温贫困水平。

#### 深度解读

该文将高温适应从单一空调可及性扩展为社会-基础设施-健康复合问题。它提示热风险治理不能只依赖能源消费扩张，否则可能加剧排放和不平等。对城市气候适应而言，遮阴、住房质量、公共冷却空间、劳动保护和电力系统韧性需要被纳入同一政策框架。

DOI: <https://doi.org/10.1038/s41893-026-01845-4>

## 植物与农业

3. Global rice paddy greenhouse gas emissions have doubled over the past six decades driven by area expansion and intensified residue incorporation

(面积扩张和秸秆还田强化推动全球稻田温室气体排放六十年翻倍)

Nature Food | 2026-05-22 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Using a data-driven model, a process-based ecosystem model and a meta-analysis of more than 1,255 field experiment sites, this study estimates that net greenhouse gas emissions from global rice paddies roughly doubled from 1961-1980 to 2001-2020. The increase was mainly driven by higher soil CO<sub>2</sub> and CH<sub>4</sub> emissions, linked to rice-area expansion and intensified residue incorporation.

#### 中文摘要

研究使用数据驱动模型、过程生态系统模型和 1,255 多个田间试验点的荟萃分析，估算全球稻田净温室气体排放从 1961-1980 年到 2001-2020 年约翻倍。增长主要来自土壤 CO<sub>2</sub> 和 CH<sub>4</sub> 排放上升，与稻田面积扩张和秸秆还田强化密切相关。

#### 深度解读

稻田既是粮食安全核心系统，也是甲烷和二氧化碳排放的重要来源。该研究的价值在于用多方法交叉约束长期排放趋势，减少单一模型偏差。未来水稻低碳管理需要在粮食产量、秸秆资源化、间歇灌溉和土壤碳管理之间寻找区域化组合，而不是简单压缩稻作。

DOI: <https://doi.org/10.1038/s43016-026-01355-8>

## 环境与气候

### 4. Transport-constrained eddy-mean energetics sustain extreme rainfall over China

(输送约束的涡旋-平均能量过程维持中国极端降雨)

Communications Earth & Environment | 2026-05-22 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Extreme summer rainfall over China has intensified in recent decades, but the mechanisms organizing moisture transport and sustained ascent remain uncertain. This open-access study links transport-constrained eddy-mean energetics to the maintenance of extreme rainfall, offering physical insight for projections and adaptation planning in flood-prone regions.

#### 中文摘要

研究关注中国夏季极端降雨增强背后的水汽输送和持续上升运动机制，提出输送约束下的涡旋-平均能量过程可维持极端降雨。该机制有助于解释长江、珠江等人口和经济密集区洪涝风险上升。

#### 深度解读

极端降雨研究常停留在水汽增多和环流异常层面，而这项工作进一步追踪能量转换与输送约束。它对提高区域气候模式和灾害预警的物理一致性有意义。对生态环境管理而言，极端降雨会通过洪涝、滑坡、污染物再悬浮和河湖营养盐输入影响流域生态安全。

DOI: <https://doi.org/10.1038/s43247-026-03547-3>

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## 海洋与水生

5. Ecological and genomic signatures of the convergent evolution of planktivory in fossil and living reef fishes over deep time

(深时尺度上化石与现生礁鱼浮游生物食性的趋同进化生态与基因组信号)

Nature Communications | 2026-05-22 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

This study examines ecological and genomic signatures of convergent evolution of planktivory in fossil and living reef fishes over deep time. By linking fossil evidence with living reef-fish genomic and ecological patterns, it provides a long-term view of how reef fish trophic strategies repeatedly emerged.

#### 中文摘要

研究结合化石和现生礁鱼资料，追踪浮游生物食性在深时尺度上的趋同演化，并从生态和基因组层面识别相关信号。该工作为理解珊瑚礁鱼类营养策略如何反复出现提供了长期演化视角。

#### 深度解读

珊瑚礁鱼类功能多样性是维持礁生态系统稳定的重要基础。把化石记录与现生基因组结合，可以区分短期生态适应和长期演化约束。该研究也提示，在气候变化和珊瑚礁退化背景下，评估功能群脆弱性需要考虑其演化可塑性和历史路径。

## 环境与气候

### 6. Onset seasonality controls compound streamflow drought risk at a global scale (起始季节性控制全球复合径流干旱风险)

npj Natural Hazards | 2026-05-21 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Streamflow droughts threaten water, food and energy security. This study defines compound streamflow droughts as low streamflow coinciding with or preceded by precipitation deficits and followed by elevated evaporative demand within two months of onset, and shows that onset seasonality is a key global control on compound drought risk.

#### 中文摘要

研究将复合径流干旱定义为低径流与降水亏缺相伴或相继出现，并在起始前后两个月内叠加较高蒸发需求的事件。结果表明，干旱起始季节性是控制全球复合径流干旱风险的重要因素。

#### 深度解读

传统干旱指标往往把降水、径流和蒸发需求分开处理，而复合干旱更接近水资源系统承受的真实压力。起始季节决定植被需水、蓄水状态和人类用水峰值，因此会放大或缓冲生态影响。流域管理需把季节性预警转化为水库调度、生态流量和农业用水的动态规则。

DOI: <https://doi.org/10.1038/s44304-026-00221-8>

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## 植物与农业

### 7. A transcriptional atlas of early Arabidopsis seed development suggests mechanisms for inter-tissue coordination (拟南芥早期种子发育转录图谱揭示组织间协调机制)

Nature Plants | 2026-05-21 | **【Latest】**



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Successful seed development in flowering plants requires coordination among the embryo, endosperm and maternal seed coat. This resource article presents a transcriptional atlas of early *Arabidopsis* seed development and suggests mechanisms for inter-tissue coordination during reproduction.

#### 中文摘要

被子植物种子发育需要胚、胚乳和母体种皮三个遗传来源不同的组织协调。该资源型研究构建拟南芥早期种子发育的转录图谱，并提出组织间协调发育的潜在机制。

#### 深度解读

虽然该文偏植物发育基础研究，但对农业生态系统中的种子产量、逆境响应和繁殖成功具有基础意义。高分辨率转录图谱可帮助识别胚乳、种皮和胚之间的信号通路。长期看，这类资源会支持作物育种中对种子大小、灌浆和环境适应性的精准调控。

DOI: <https://doi.org/10.1038/s41477-026-02295-8>

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## 环境与气候

### 8. Warming erodes climate connectivity for terrestrial vertebrates

(变暖削弱陆生脊椎动物的气候连通性)

Nature Climate Change | 2026-05-18 | 【Recent】



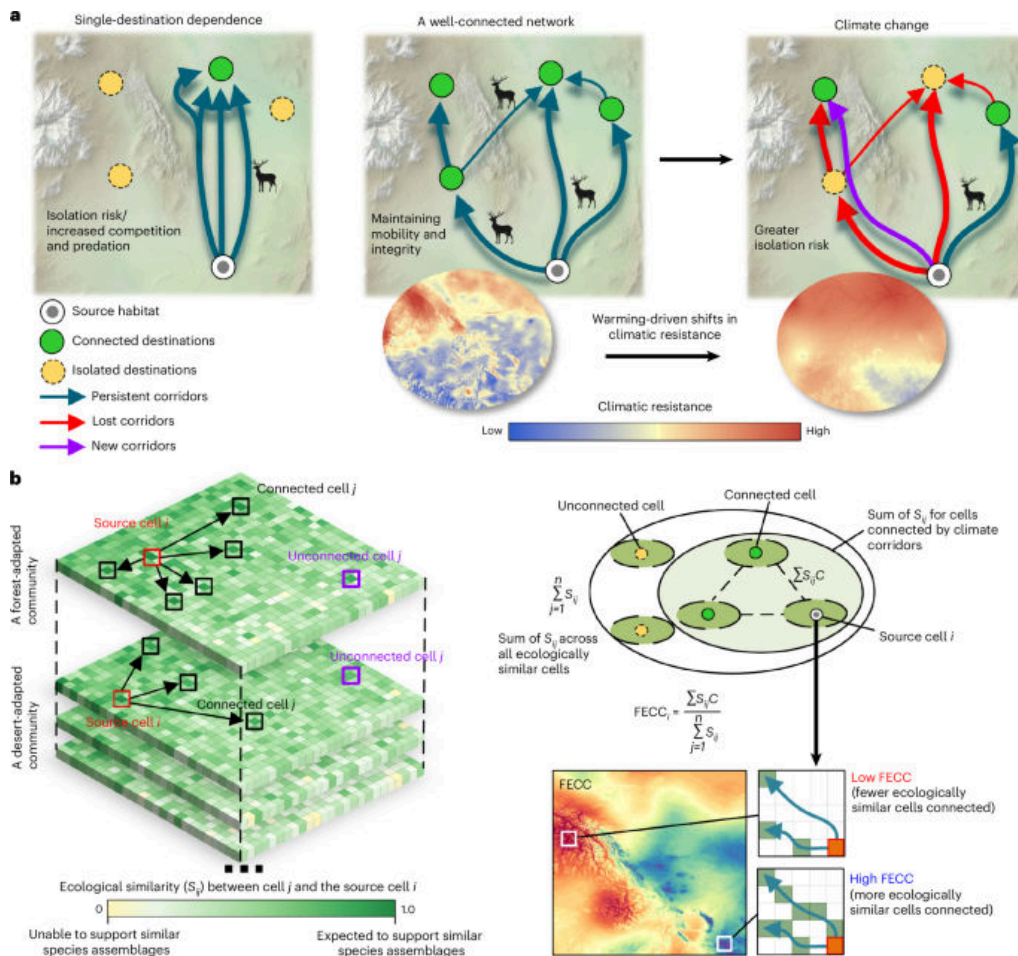


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Climate Change, Published online: 18 May 2026; doi:10.1038/s41558-026-02658-1 The authors consider connectivity—specifically, the functional effectiveness of climate connectivity (FECC)—under climate change. Under high emissions (SSP5-8.5), FECC is projected to decline across 77% of the global land area by 2061–2080, risking ecological isolation and subsequent biodiversity loss.

## 中文摘要

作者从气候连通性的功能有效性（FECC）出发，评估气候变化下陆生脊椎动物沿适宜气候通道迁移的可能性。在高排放情景 SSP5-8.5 下，到 2061—2080 年，全球约 77% 陆地区域的 FECC 预计下降，增加生态隔离和生物多样性损失风险。

## 深度解读

这项研究把保护区和栖息地连通性从静态空间格局推进到动态气候适宜性层面。其核心意义在于指出，仅保留现有栖息地斑块并不足以保障物种随气候迁移。对保护规划而言，廊道设计需要优先考虑未来气候通道的持续性，而非只依据当前物种分布。

DOI: <https://doi.org/10.1038/s41558-026-02658-1>

## 海洋与水生

### 9. Crop-climate models need more realistic representations of irrigation under warming

(作物—气候模型需要更真实地表征变暖下的灌溉作用)

Nature Water | 2026-05-05 | 【Recent】

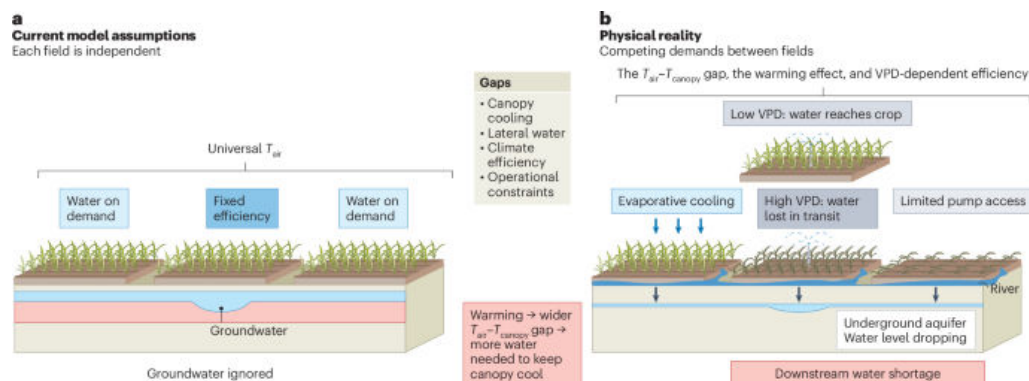


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Water, Published online: 05 May 2026; doi:10.1038/s44221-026-00642-9 Crop models increasingly project that irrigation can offset agricultural losses from climate change, informing major adaptation investments worldwide. Yet these models may systematically overestimate irrigation's protective capacity because the representation of how irrigation alters crop thermal environments remains incomplete in most large-scale assessments and assumes static efficiency under changing atmospheric conditions. More realistic assessments are urgently needed to avoid underestimating future water demands and regional adaptation failures.

## 中文摘要

作物模型越来越多地认为灌溉可抵消气候变化造成的农业损失，并影响全球适应投资。但多数大尺度评估对灌溉如何改变作物热环境的刻画仍不完整，且常假设灌溉效率在变化的大气条件下保持不变，可能系统性高估其保护能力。

## 深度解读

该文提醒气候适应评估不能把灌溉视为简单、稳定的减灾按钮。变暖会改变蒸散需求、作物冠层温度和水分利用效率，从而影响灌溉的真实收益。若模型低估未来需水量，水资源配置和农业适应政策可能面临区域性失灵。

DOI: <https://doi.org/10.1038/s44221-026-00642-9>

## 环境与气候

### 10. Seabird range contraction and dispersal under climate change

(气候变化下海鸟分布范围收缩与扩散距离增加)

Nature Climate Change | 2026-05-19 | 【Recent】

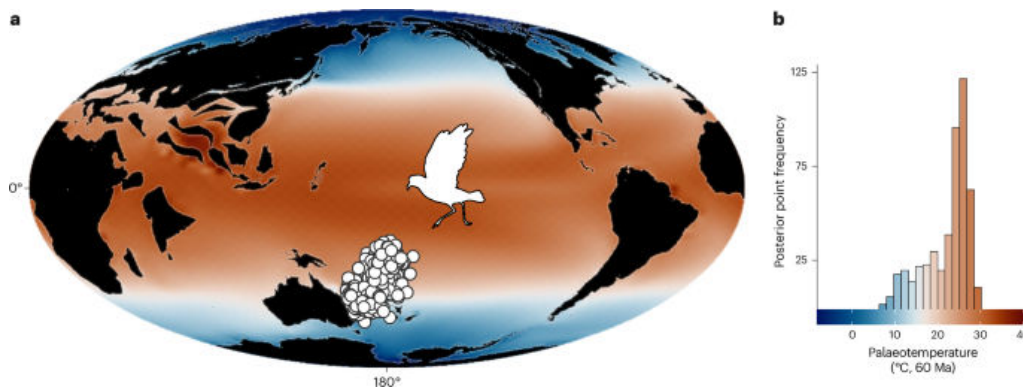


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Climate Change, Published online: 19 May 2026; doi:10.1038/s41558-026-02655-4The authors reconstruct historical seabird dispersal routes, showing that birds responded to temperature shifts by changes in range size rather than body mass. These trends are projected to persist, with higher rates of warming causing greater range contractions and longer dispersal distances.

## 中文摘要

作者重建了历史海鸟扩散路线，发现海鸟对温度变化的响应主要表现为分布范围大小变化，而不是体重变化。模型预计这一趋势将延续，升温速率越高，分布范围收缩越明显，扩散距离也越长。

## 深度解读

海鸟具有较强移动能力，但这并不意味着它们能轻松逃避气候压力。研究显示，温度变化会重塑适宜栖息地的空间结构，使种群面临更远距离扩散和繁殖地丧失的双重压力。对海洋保护区而言，固定边界可能难以覆盖未来关键栖息地。

DOI: <https://doi.org/10.1038/s41558-026-02655-4>

## 环境与气候

11. Social inequalities mediate temperature–child maltreatment associations in Africa

(社会不平等调节非洲温度与儿童虐待之间的关联)

Nature Climate Change | 2026-05-15 | 【Recent】

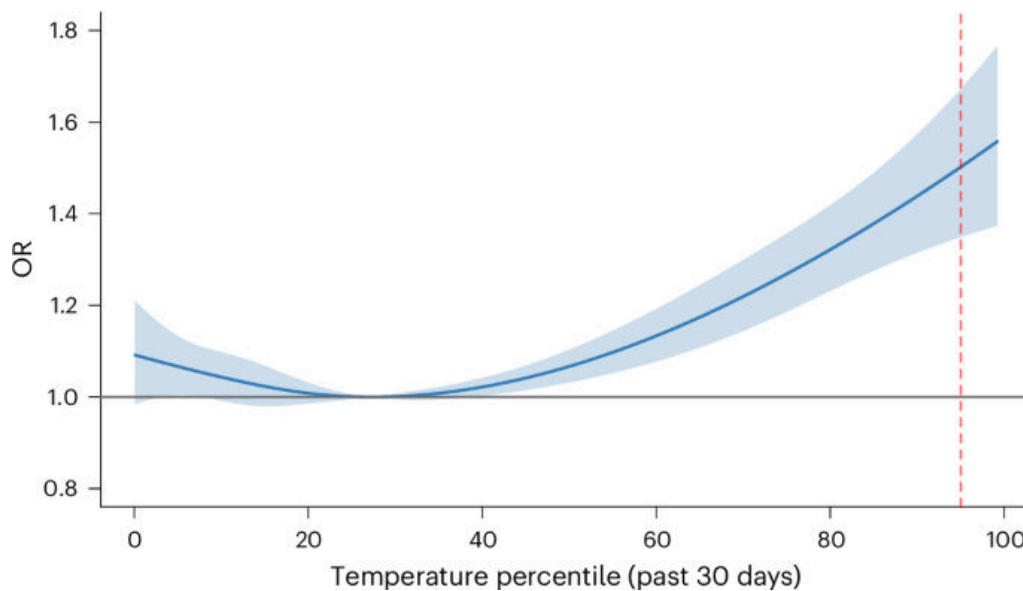


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Climate Change, Published online: 15 May 2026; doi:10.1038/s41558-026-02650-9The link between temperature and child maltreatment in Africa remains unexplored. This study demonstrates a substantial association, particularly among socio-economically disadvantaged families, driven by behavioural changes, occupational exposure and reduced household resources.

#### 中文摘要

该研究探讨非洲地区温度与儿童虐待之间此前较少被关注的联系，发现二者存在显著关联，且在社会经济弱势家庭中更为突出。可能机制包括行为变化、职业暴露增加以及家庭资源减少。

#### 深度解读

这篇文章把气候影响从生态和经济损失扩展到家庭与儿童福祉层面。高温并非均匀地作用于社会，而是通过劳动环境、收入压力和照护资源放大既有不平等。气候适应政策因此需要与社会保护、儿童保护和劳动保障联动。

DOI: <https://doi.org/10.1038/s41558-026-02650-9>

## 环境与气候

12. Future changes in seasonal sea-level variability could reshape coastal ecosystems

(未来季节性海平面变化或重塑海岸生态系统)

Nature Climate Change | 2026-05-13 | **【Recent】**



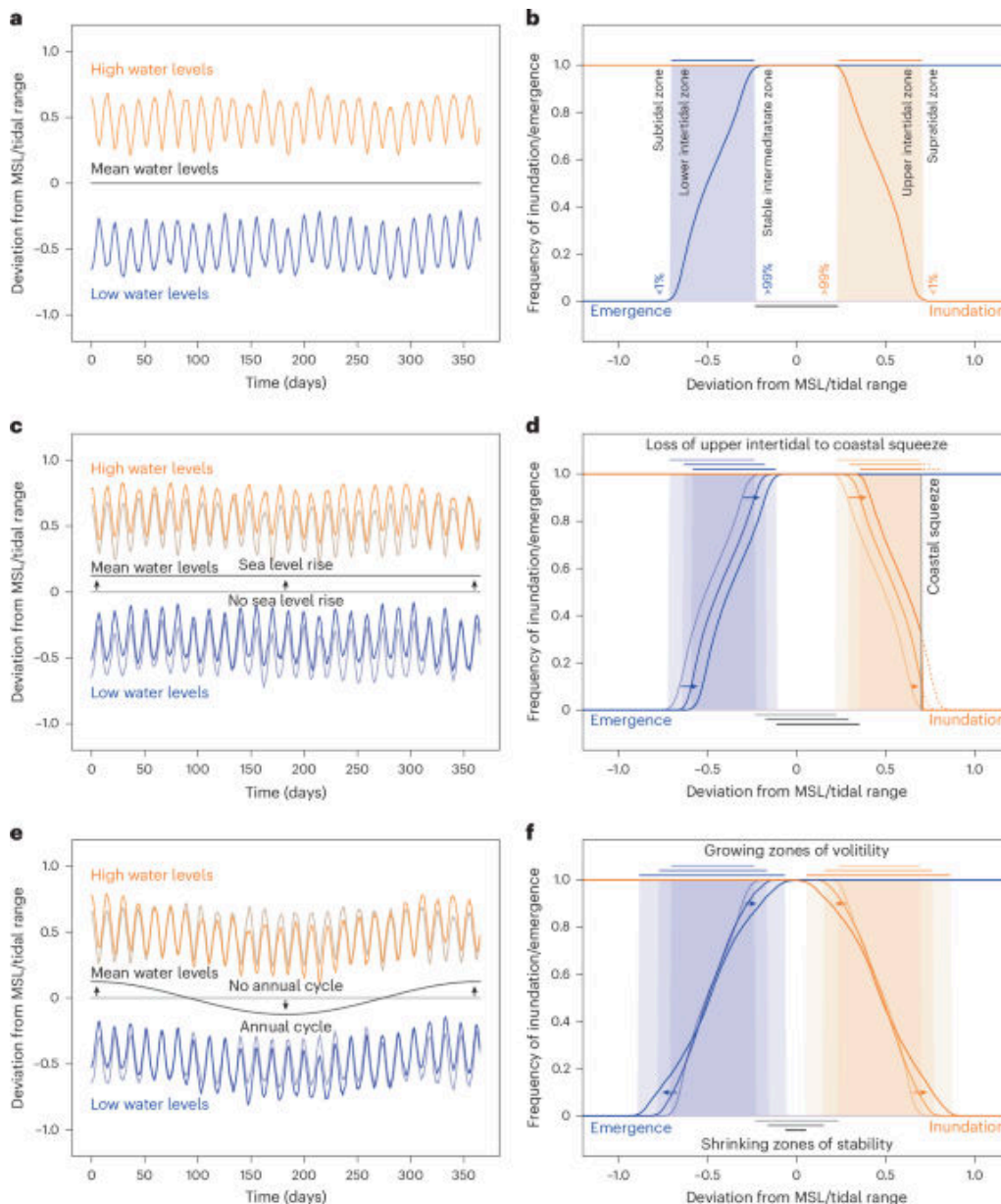


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Climate Change, Published online: 13 May 2026; doi:10.1038/s41558-026-02631-y Assessments of coastal ecosystem resilience typically consider the impacts of annual mean sea-level rise, while increases in the seasonal sea-level cycle could also affect intertidal ecosystems. The authors show how such increases can threaten intertidal zones through altering the frequency and duration of inundation and emergence events.

## 中文摘要

海岸生态系统韧性评估通常关注年平均海平面上升,但季节性海平面循环增强也可能影响潮间带生态系统。作者指出,季节性变化增加会改变淹没与出露事件的频率和持续时间,从而威胁潮间带。

## 深度解读

该研究强调海平面变化的时间结构与平均升幅同样重要。潮间带生物依赖周期性淹没和暴露维持生理与生态过程,季节节律改变会影响物种组成和栖息地边界。海岸管理需要把季节性风险纳入湿地恢复、岸线规划和生态补偿设计。

DOI: <https://doi.org/10.1038/s41558-026-02631-y>

## 环境与气候

### 13. Evapotranspiration declines prolonged by deforestation and fire in South American biomes

(南美生物群系中森林砍伐与火灾延长蒸散下降过程)

Nature Geoscience | 2026-05-19 | 【Recent】

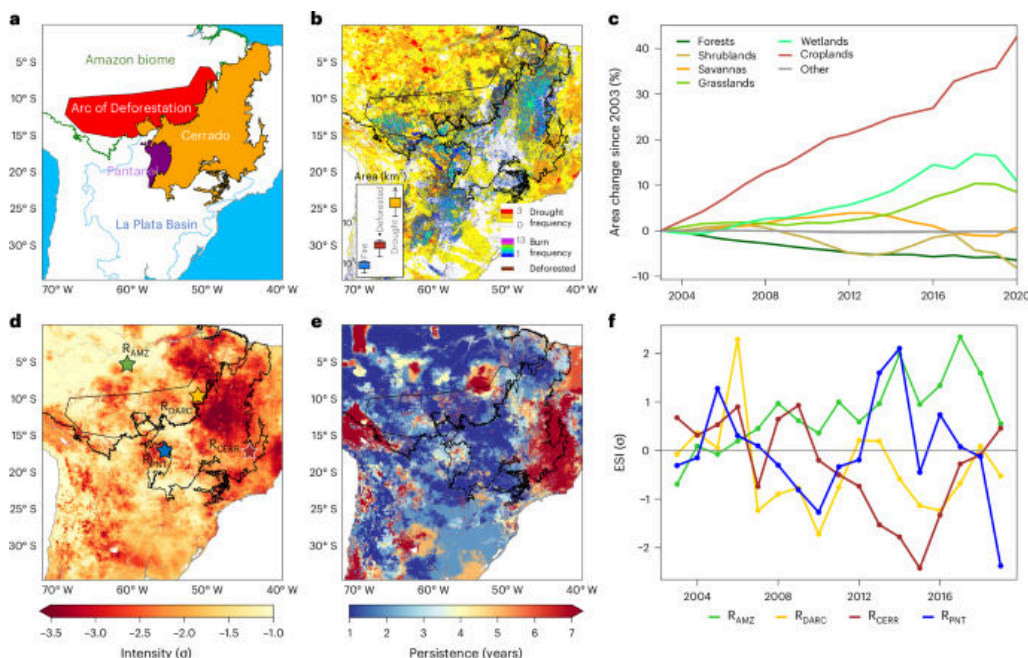


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Geoscience, Published online: 19 May 2026; doi:10.1038/s41561-026-01981-8 Human-driven disturbances intensify and prolong evapotranspiration declines across South American ecosystems, disrupting the water balance and threatening ecological resilience, according to hydrological modelling and remote sensing data.

#### 中文摘要

基于水文模型和遥感数据,研究表明人类驱动的扰动会加剧并延长南美生态系统的蒸散下降,扰乱区域水量平衡并威胁生态韧性。森林砍伐和火灾是影响植被—水循环耦合的重要压力源。

#### 深度解读

蒸散是陆地生态系统调节区域气候和水循环的关键通量。扰动后蒸散长期偏低意味着植被恢复、降水反馈和热环境调节都可能受到持续影响。该结果支持将控火、减少毁林和生态恢复作为水安全政策的一部分。

DOI: <https://doi.org/10.1038/s41561-026-01981-8>

## 环境与气候

### 14. Reshaping river flow

(重塑河流流动格局)



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Geoscience, Published online: 13 May 2026; doi:10.1038/s41561-026-01993-4 Earth's landmass has been sculpted by rivers for millions of years. Humans are now reshaping these landscapes as engineered modifications and the impacts of anthropogenic climate change alter river connectivity, water resources, and sediment transport.

#### 中文摘要

文章指出，河流在数百万年中塑造了陆地地貌，而人类活动正在通过工程改造和人为气候变化改变河流连通性、水资源格局与泥沙输移。该文属于对河流系统人类重塑趋势的概述性讨论。

#### 深度解读

河流变化不只是水量问题，也涉及地貌演化、沉积物供给和生态连通性。水坝、河道整治与气候变化共同作用，会改变河流维持三角洲、洪泛平原和栖息地的能力。未来流域管理需要从单一工程效率转向河流过程完整性。

DOI: <https://doi.org/10.1038/s41561-026-01993-4>

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## 保护与生物多样性

15. Ecological and climatic drivers of wildlife road mortality in Kenya's Athi-Kapiti Plains

(肯尼亚 Athi-Kapiti 平原野生动物道路死亡的生态与气候驱动因素)

Biodiversity and Conservation | 2026-05-16 | 【Recent】



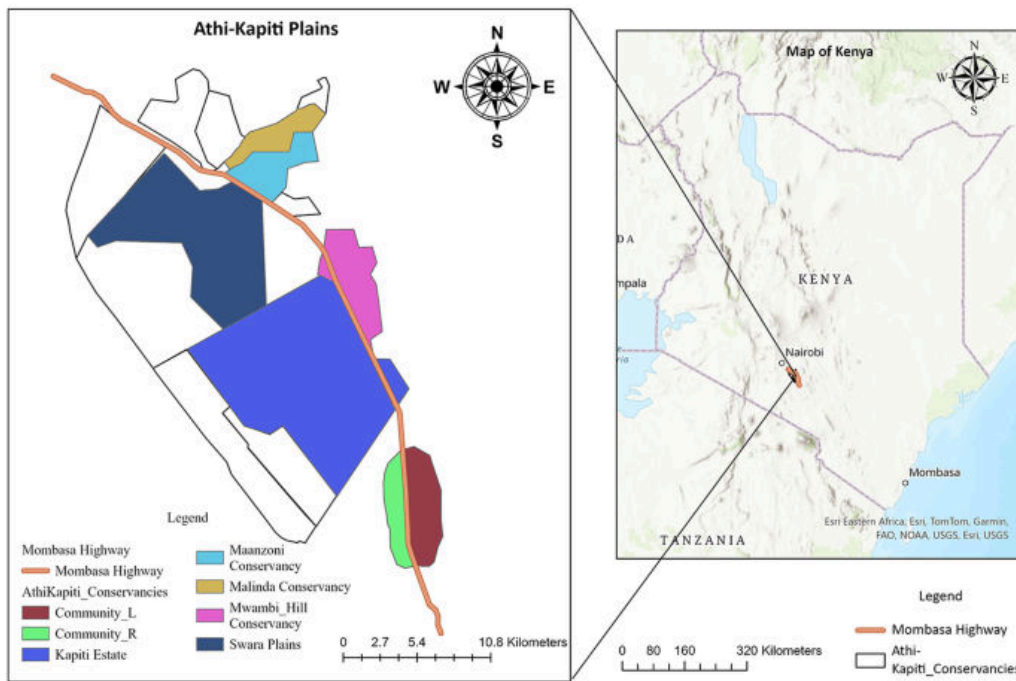


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1007%2F>

## Abstract

Roads and habitat fragmentation pose significant conservation challenges for many animals, yet the environmental factors that influence roadkill risk remain poorly understood. In Kenya, the Nairobi–Mombasa Highway cuts through the ranges of numerous wildlife species within one of the country’s most biodiverse regions, the Athi-Kapiti Plains. To investigate the environmental factors influencing roadkill risk in this region, we combined remote sensing data on long-term drought (Standardized Precipitation Evapotranspiration Index, SPEI) and vegetation greenness (Normalized Difference Vegetation Index, NDVI) with 6,240 km of road transect surveys conducted in 2023 and historical roadkill records from 2020 to 2022. We recorded 218 wildlife roadkills involving 11 species, predominantly plains ze

## 中文摘要

研究关注内罗毕—蒙巴萨公路穿越肯尼亚 Athi-Kapiti 平原所造成的道路致死风险。作者结合长期干旱指数、植被绿度遥感数据、2023 年 6240 公里道路样线调查和 2020—2022 年历史记录，分析影响不同野生动物道路死亡的环境因素。

## 深度解读

道路生态风险具有明显的时空异质性，受干旱、植被状态和动物移动需求共同影响。该研究把道路死亡从单纯交通问题转化为气候—栖息地—动物行为耦合问题。结果可用于优化限速区、野生动物通道和热点路段监测。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03361-5>

## 保护与生物多样性

16. Land use around protected areas affects functional diversity of anuran assemblages in ponds of Southern Brazilian Highland Grasslands

(保护地周边土地利用影响巴西南部高地草原池塘蛙类群落功能多样性)

Biodiversity and Conservation | 2026-05-15 | 【Recent】



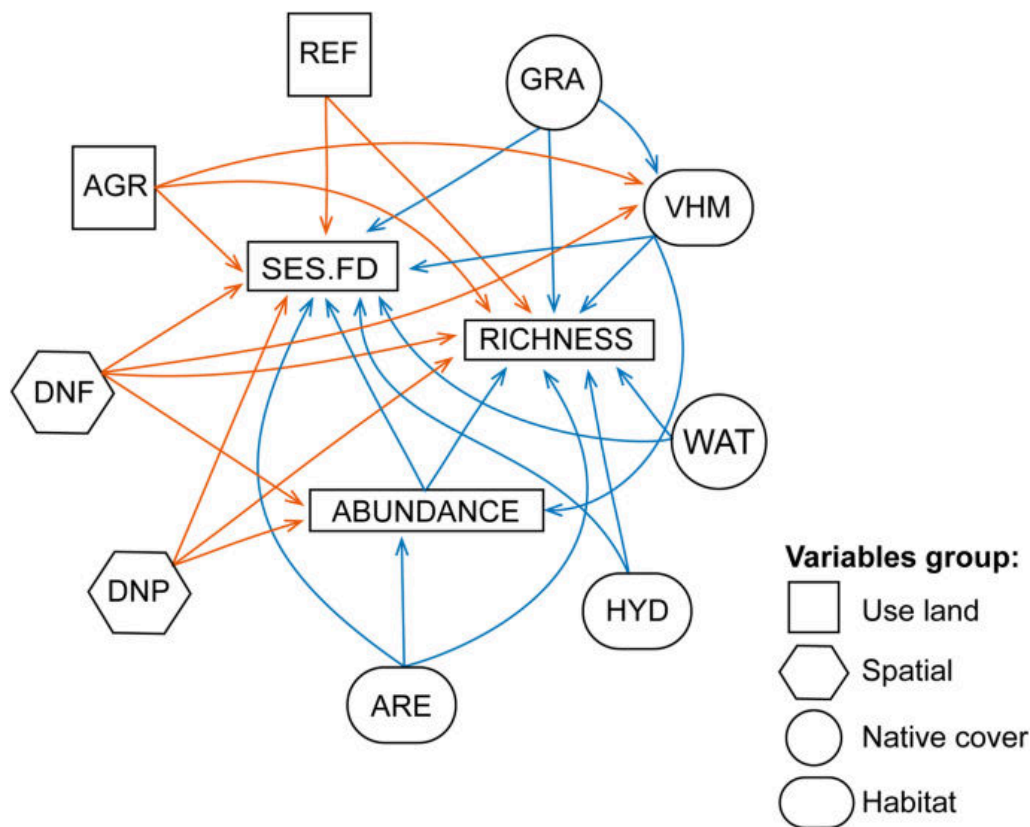


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1007%2F>

#### Abstract

Anthropogenic land-use change drives biodiversity loss by altering habitat structure, reducing connectivity, and reshaping ecological processes, even within protected areas. Although protected areas are key for biodiversity conservation, their effectiveness may depend on the surrounding landscape context. This study evaluates the effects of land use around and within protected areas on species richness and functional diversity in amphibian assemblage. Data were collected during 12 monthly expeditions between January 2013 and December 2014, sampling 81 ponds across eight protected areas and surrounding landscapes in southern Brazil. We measured species richness, abundance, and 21 ecomorphological traits, using phylogenetic eigenvector regression to control phylogenetic autocorrelation. Func

#### 中文摘要

该研究评估保护地内外土地利用对两栖动物物种丰富度和功能多样性的影响。作者在巴西南部 8 个保护地及周边景观的 81 个池塘开展 12 次月度调查，测量物种丰富度、丰度和 21 项生态形态性状，并控制系统发育自相关。

#### 深度解读

保护地成效并不只取决于边界内部管理，周边景观同样会影响池塘生境和蛙类功能结构。功能多样性比物种数更能反映群落对环境过滤和生态过程变化的响应。该研究支持在保护地规划中纳入缓冲区土地利用和湿地网络连通性。

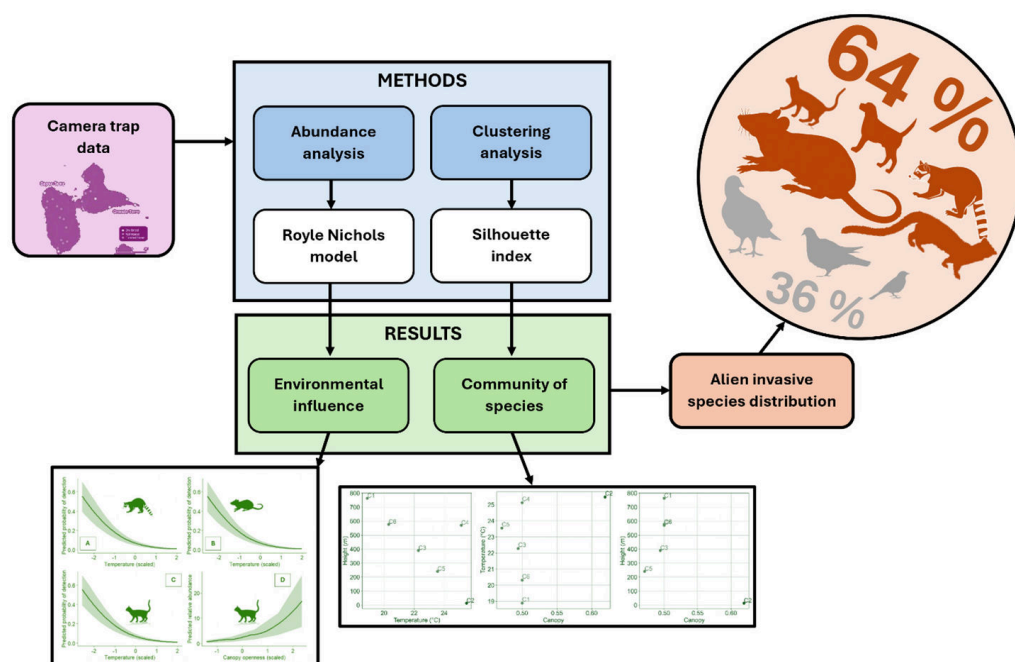
DOI: <https://link.springer.com/article/10.1007/s10531-026-03358-0>

## 综合顶刊

17. Accelerated Himalayan river meandering and dynamics due to climate change

(气候变化导致喜马拉雅河流弯曲与动力过程加速)

Science | 2026-05-14T07:00:00Z | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

该 Science 论文题名显示，研究关注气候变化背景下喜马拉雅河流曲流发育和河道动力过程的加速变化。由于所给摘要为空，具体方法、区域和定量结论需以原文为准。

深度解读

喜马拉雅河流连接冰雪融水、季风降水、泥沙输移和下游洪水风险，是气候变化影响高度敏感的地貌系统。若河道摆动和迁移加速，将直接影响河岸侵蚀、基础设施安全和洪泛平原管理。该主题提示高山河流研究需要把气候驱动与河道形态动力学结合起来。

DOI: <https://doi.org/10.1126/science.adg8401>

## 生态学核心

18. Larger forest patches have greater per-area productivity

(较大的森林斑块具有更高的单位面积生产力)

Nature Ecology & Evolution | 2026-05-15 | 【Recent】

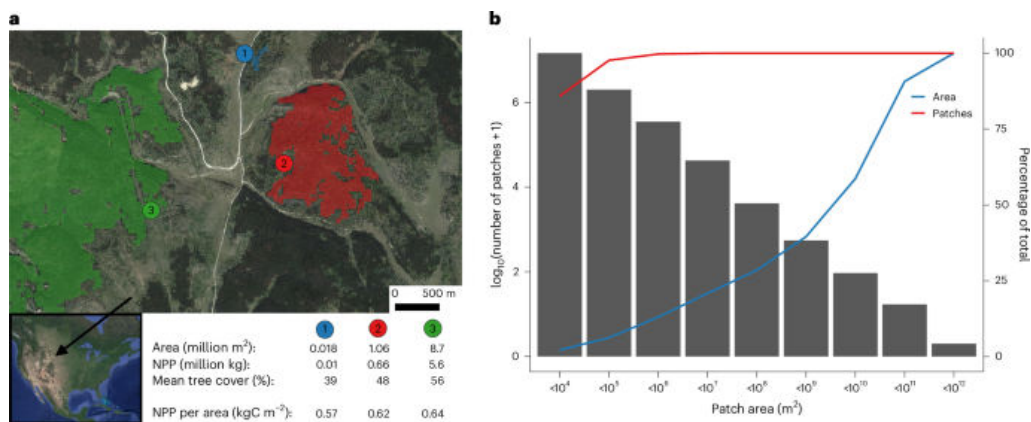


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Ecology & Evolution, Published online: 15 May 2026; doi:10.1038/s41559-026-03075-5 Using remote sensing data, spatial modelling and counterfactual scenarios, this analysis reveals a positive relationship between forest patch size and productivity. This implies that preserving many small forest patches may not offset the carbon losses from larger contiguous forests.

## 中文摘要

研究利用遥感数据、空间建模和反事实情景，发现森林斑块大小与生产力之间存在正相关关系。这意味着保存许多小型森林斑块，可能无法抵消大型连续森林丧失带来的碳损失。

## 深度解读

该结论挑战了只以面积总量衡量森林保护成效的做法。大型连续森林可能通过边缘效应较弱、微气候更稳定和生态过程更完整而维持更高生产力。碳汇政策应避免把破碎小斑块简单等同于完整森林。

DOI: <https://doi.org/10.1038/s41559-026-03075-5>

## 环境与气候

### 19. Personal experiences matter for climate action

(个人经历对气候行动至关重要)

Nature Climate Change | 2026-05-19 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Climate Change, Published online: 19 May 2026; doi:10.1038/s41558-026-02640-x Climate change impacts are no longer distant but have entered people's everyday experiences. Here we look back on a 2011 paper that showed how direct personal experience shapes people's climate change perceptions, their beliefs about the efficacy of their action and willingness to act, and how the field of research has evolved.

#### 中文摘要

文章回顾一篇 2011 年研究，讨论直接个人经历如何塑造公众对气候变化的感知、对自身行动有效性的信念以及行动意愿。随着气候影响进入日常生活，该领域研究也在持续发展。

#### 深度解读

气候传播不再只是传递抽象风险，而要了解人们如何从亲身经历中形成判断。个人经历能提高风险感知，但是否转化为行动还取决于效能感、社会规范和可行选择。该文对气候沟通和社区适应动员具有现实意义。

DOI: <https://doi.org/10.1038/s41558-026-02640-x>

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## 土壤与生物地球化学

20. The efficiency and ocean acidification mitigation potential of ocean alkalinity enhancement on multi-centennial timescales

(多世纪尺度上海洋碱度增强的效率及缓解海洋酸化潜力)

Biogeosciences | Mon, 18 May 2026 02:09:04 +0200 | **【Recent】**





图：Graphical abstract / article figure source:

#### Abstract

The efficiency and ocean acidification mitigation potential of ocean alkalinity enhancement on multi-centennial timescales Hendrik Grosselindemann, Friedrich A. Burger, and Thomas L. Frölicher *Biogeosciences*, 23, 3299–3321, <https://doi.org/10.5194/bg-23-3299-2026>, 2026 We assess the long-term carbon cycle, climate and ocean acidification response of ocean alkalinity enhancement (OAE) using an emission-driven Earth system model across warming stabilization scenarios. OAE lowers atmospheric CO<sub>2</sub> and produces a scenario-independent linear cooling. Gross carbon capture efficiency remains high, while net efficiencies decline over time due to carbon-cycle feedbacks. Ocean acidification mitigation by OAE is dominated by CO<sub>2</sub> drawdown on long timescales.

#### 中文摘要

研究使用排放驱动的地球系统模型，在不同变暖稳定情景下评估海洋碱度增强（OAE）对长期碳循环、气候和海洋酸化的影响。结果显示，OAE 可降低大气 CO<sub>2</sub> 并产生相对情景独立的线性降温，总碳捕获效率较高，但净效率会因碳循环反馈随时间下降；长期尺度上海洋酸化缓解主要由 CO<sub>2</sub> 抽降主导。

#### 深度解读

该研究把 OAE 从短期碳移除概念推进到多世纪反馈评估。总效率与净效率的差异说明，海洋碳汇工程必须考虑再平衡和反馈过程，而不能只看初始吸收。其结论有助于约束 OAE 的气候收益、酸化缓解收益和长期治理风险。

DOI: <https://doi.org/10.5194/bg-23-3299-2026>

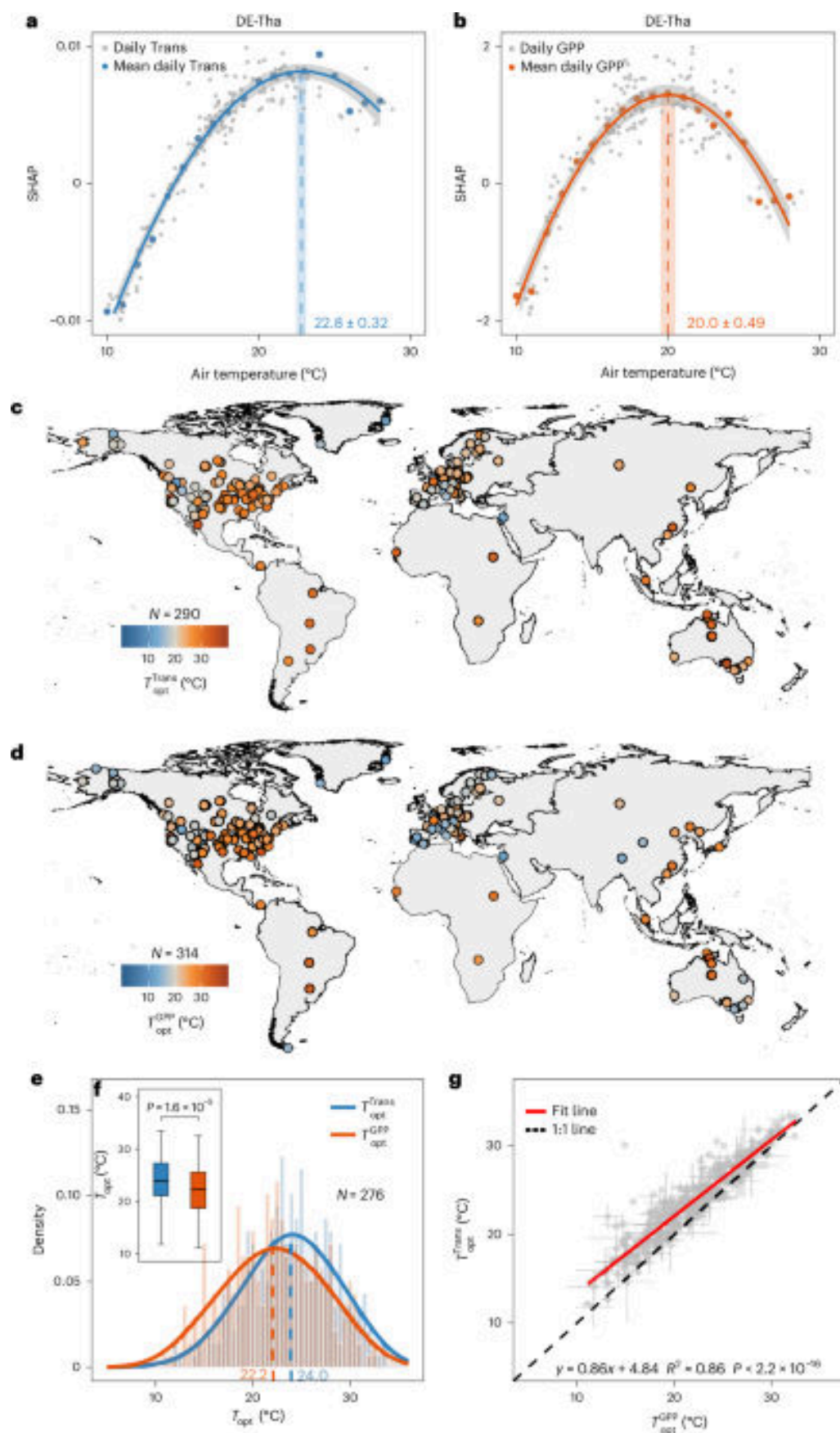
## 综合顶刊

21. Historical legacies shape continental variation in contemporary mammal food webs

(历史遗产塑造当代哺乳动物食物网的大陆差异)

PNAS | 2026-04-27T07:00:00Z | 【Recent】





图：Graphical abstract / article figure source:

**Significance**Trophic interactions underpin ecosystem structure and function, yet how past species losses, evolutionary histories, and environmental variability shape macroecological patterns of food webs remains poorly understood. We present continental-...

该 PNAS 文章指出, 营养相互作用支撑生态系统结构与功能, 但过去物种丧失、进化历史和环境变异如何塑造食物网宏生态格局仍认识不足。研究从大陆尺度分析当代哺乳动物食物网差异及其历史来源。

食物网不是只由当前环境决定，也带有深刻的历史遗产。过去灭绝和谱系演化会改变可形成的捕食—被捕食关系，从而影响现代生态系统功能。该视角有助于解释不同大陆在恢复、重引入和生态功能重建中的差异。

# 土壤与生物地球化学

(北海与瓦登海交界处从潮汐到季节尺度的海洋碳酸盐体系变异)

**a** Phylogenetic tree and genomic feature heatmap. The tree shows relationships between various brown algae species, including *Ecotocarpus* sp. 7, *Ecotocarpus aliculosus*, *Ecotocarpus crocuniorum*, *Ecotocarpus fasciculatus*, *Pleurocladia laevis*, *Scytosiphon promiscuus*, *Platylella littoralis*, *Chordaria linearis*, *Portierinema fluviatile*, *Undaria pinnatifida*, *Saccharina latissima*, *Desmarestia dudresnayi*, *Desmarestia herbacea*, *Saccorhiza polyschides*, *Fucus serratus*, *Ascophyllium nodosum*, *Dichotria dichotoma*, *Schizocladia ischiensis*, *Tribonema minus*, *Chrysoparadoxa australica*, *Heterosigma akashiwo*, *Fistulifera solaris*, *Phaeodactylum tricornutum*, *Chondrus crispus*, *Cyanidioschyzon merolae*, *Homo sapiens*, *Mus musculus*, *Drosophila melanogaster*, *Saccharomyces cerevisiae*, *Arabidopsis thaliana*, *Oryza sativa*, *Marchantia polymorpha*, *Chlamydomonas reinhardtii*, and *Cyanophora paradoxa*. The heatmap displays methylation levels (DNA), PRC1 and PRC2 occupancy, SET domain proteins, and DOT1 occupancy across the genome.

**b** Representative images of brown algae species: *Ecotocarpus* sp. 7, *Scytosiphon promiscuus*, *Undaria pinnatifida*, *Desmarestia herbacea*, *Desmarestia dudresnayi*, *Schizocladia ischiensis*, and *Ecotocarpus* sp. 7 (scale bar 100  $\mu$ m).

**c** Genomic complexity and sex dimorphism. The heatmap shows complexity (number of cell types) and sex dimorphism (dioecious/monocious) for various species. The color scale ranges from 0 (white) to 700 (dark red) for complexity and from 0 (white) to 100 (dark blue) for sex dimorphism.

**d** Mass spectrometry, Western blot, and ChIP-seq results. The heatmap shows the detection of various proteins across different species. The color scale ranges from 0 (white) to 100 (dark blue) for mass spectrometry and Western blot, and from 0 (white) to 100 (dark blue) for ChIP-seq.

23

## Abstract

Marine carbonate system variability from tidal to seasonal timescales at the interface between the North Sea and Wadden Sea Yasmina Ourradi, Gert-Jan Reichart, Sonja van Leeuwen, and Matthew P. Humphreys Biogeosciences, 23, 3323–3346, <https://doi.org/10.5194/bg-23-3323-2026>, 2026 Coastal seas play a key role in the global carbon cycle, yet what drives their carbonate chemistry remains poorly understood. We measured pH at high frequency for a year at the Wadden Sea-North Sea interface. Biological activity drives pH changes, while freshwater inflow shapes total alkalinity. Both processes influence dissolved inorganic carbon. The Wadden Sea acts as a net carbon dioxide source to the atmosphere. High-frequency monitoring is essential under a changing climate.

## 中文摘要

研究在瓦登海—北海交界处进行一年高频 pH 观测，分析海岸海域碳酸盐化学的驱动因素。结果表明，生物活动主导 pH 变化，淡水输入影响总碱度，两者共同影响溶解无机碳；瓦登海表现为大气 CO<sub>2</sub> 的净源。

## 深度解读

海岸海域碳循环具有强烈的高频波动，低频采样容易遗漏关键过程。生物生产、呼吸和淡水输入共同控制碳酸盐体系，使酸化风险呈现潮汐到季节尺度的变化。该研究凸显在气候变化下建立高频监测网络的必要性。

DOI: <https://doi.org/10.5194/bg-23-3323-2026>

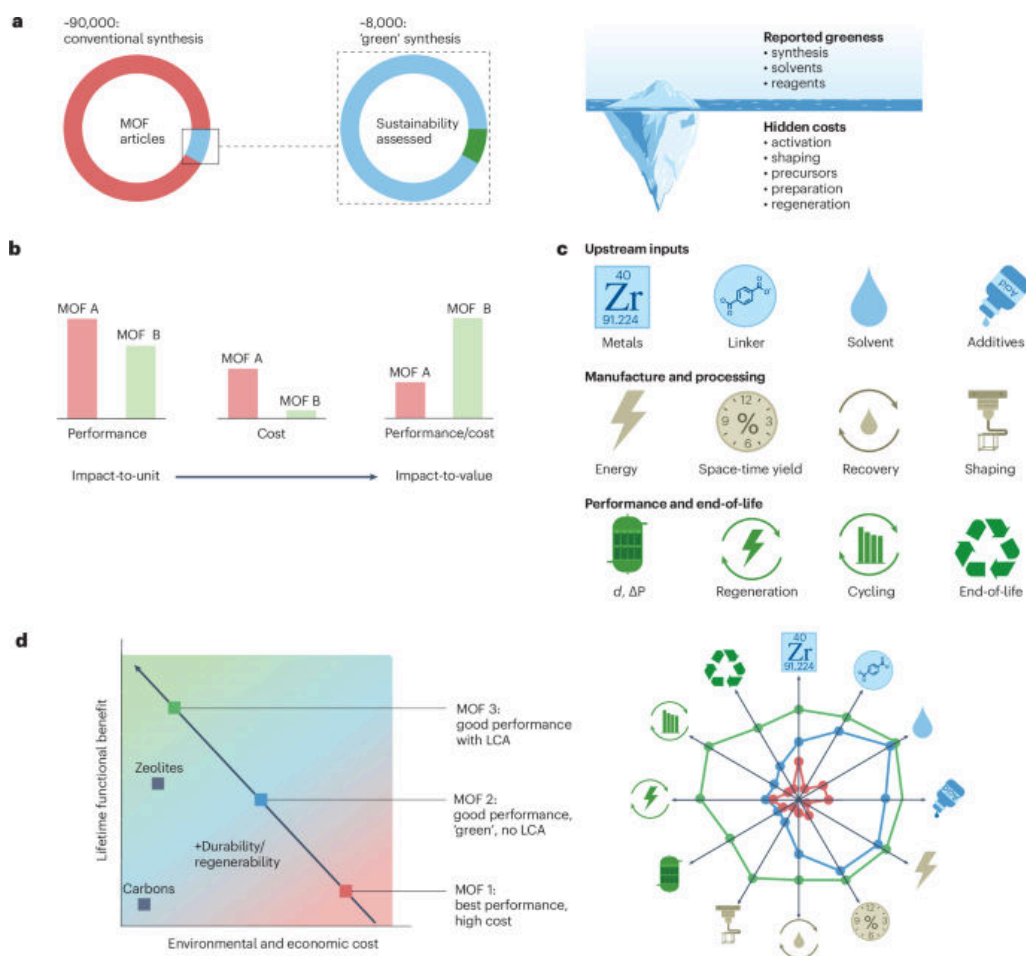
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## 环境与气候

23. Towards sustainable metal–organic frameworks from lab to practice

(推动金属—有机框架材料从实验室走向实践的可持续性评估)

Nature Sustainability | 2026-05-19 | **【Recent】**



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Sustainability, Published online: 19 May 2026; doi:10.1038/s41893-026-01839-2 Realistic life-cycle impacts of metal–organic frameworks (MOFs) are rarely quantified, despite their promise for carbon capture, water treatment, catalysis and other environmental applications. Here we propose a compact, actionable sustainability toolkit for MOFs to systematically verify claims of ‘green’.

## 中文摘要

金属–有机框架材料（MOFs）在碳捕集、水处理和催化等环境应用中具有潜力，但其真实生命周期影响很少被量化。作者提出一套紧凑、可操作的 MOF 可持续性工具包，用于系统核验“绿色”声明。

## 深度解读

新材料的环境价值不能只由单项性能指标决定，还必须纳入原料、合成、能耗、寿命和末端处置。MOF 若要规模化应用，需要证明其全生命周期收益大于替代技术。该文为材料研发早期嵌入可持续性筛选提供了方法框架。

DOI: <https://doi.org/10.1038/s41893-026-01839-2>

## 土壤与生物地球化学

24. Relative uptake of carbonyl sulphide to carbon dioxide: insights from a coupled boundary layer–canopy inverse modelling framework



(羰基硫相对于二氧化碳吸收的比例：耦合边界层—冠层反演模型的启示)

Biogeosciences | Mon, 11 May 2026 02:09:04 +0200 | 【Recent】



图：Graphical abstract / article figure source:

#### Abstract

Relative uptake of carbonyl sulphide to carbon dioxide: insights from a coupled boundary layer – canopy inverse modelling framework Peter J. M. Bosman, Maarten C. Krol, Laurens N. Ganzeveld, Felix M. Spielmann, and Georg Wohlfahrt Biogeosciences, 23, 3225–3252, <https://doi.org/10.5194/bg-23-3225-2026>, 2026 Carbonyl sulphide (COS) is a trace gas that can be used to estimate plant CO<sub>2</sub> uptake. For this, the ratio of deposition velocities of COS and CO<sub>2</sub> (leaf relative uptake - LRU) is relevant. We use a soil –canopy –atmospheric mixed layer model to simulate COS and CO<sub>2</sub> plant uptake in needleleaf ecosystems, and derive LRU. We find significant in-canopy variability of LRU, and develop a regression model for canopy-scale LRU. The results can contribute to improving COS-based ecosystem plant C

#### 中文摘要

羰基硫 (COS) 可用于估算植物 CO<sub>2</sub> 吸收，其中 COS 与 CO<sub>2</sub> 沉降速度之比，即叶片相对吸收量 (LRU)，是关键参数。研究利用土壤—冠层—大气混合层模型模拟针叶林生态系统中 COS 和 CO<sub>2</sub> 的植物吸收，发现冠层内部 LRU 具有显著变异，并建立冠层尺度 LRU 回归模型。

#### 深度解读

COS 示踪法为独立约束陆地光合作用提供了重要途径，但其可靠性依赖对冠层过程的精细刻画。冠层内部光照、气孔导度和湍流结构差异会使 LRU 并非固定常数。该研究有助于改进基于 COS 的生态系统碳吸收估算。

DOI: <https://doi.org/10.5194/bg-23-3225-2026>

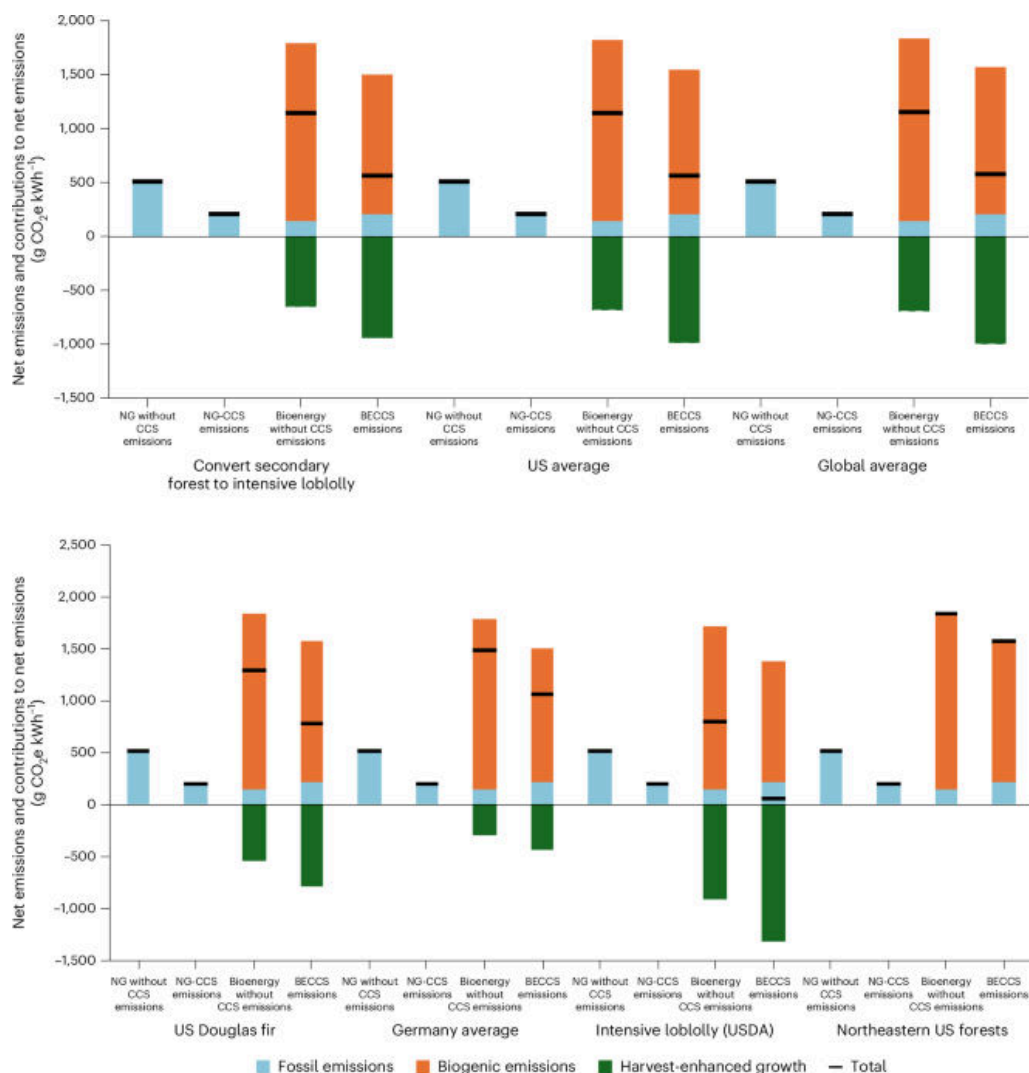
## 生态学核心

25. Tree Species Diversity Suppresses Soil Carbon Priming Effects in a Subtropical Forest



## (树种多样性抑制亚热带森林土壤碳激发效应)

Ecology Letters | Sun, 17 May 2026 06:05:30 -0700 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

### 中文摘要

该 Ecology Letters 论文题名表明，研究关注亚热带森林中树种多样性对土壤碳激发效应的影响。由于所给摘要为空，具体实验设计、机制和效应大小需以原文为准。

### 深度解读

土壤碳激发效应决定新鲜碳输入是否会加速旧碳分解，是森林碳汇稳定性的关键机制。树种多样性可能通过凋落物质量、根系输入和微生物群落组成改变这一过程。该主题对解释混交林碳固持优势和森林经营策略具有重要意义。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70400>

## 生态学核心

26. Legal compensation values for fauna species across Europe reveal taxonomic and conservation biases

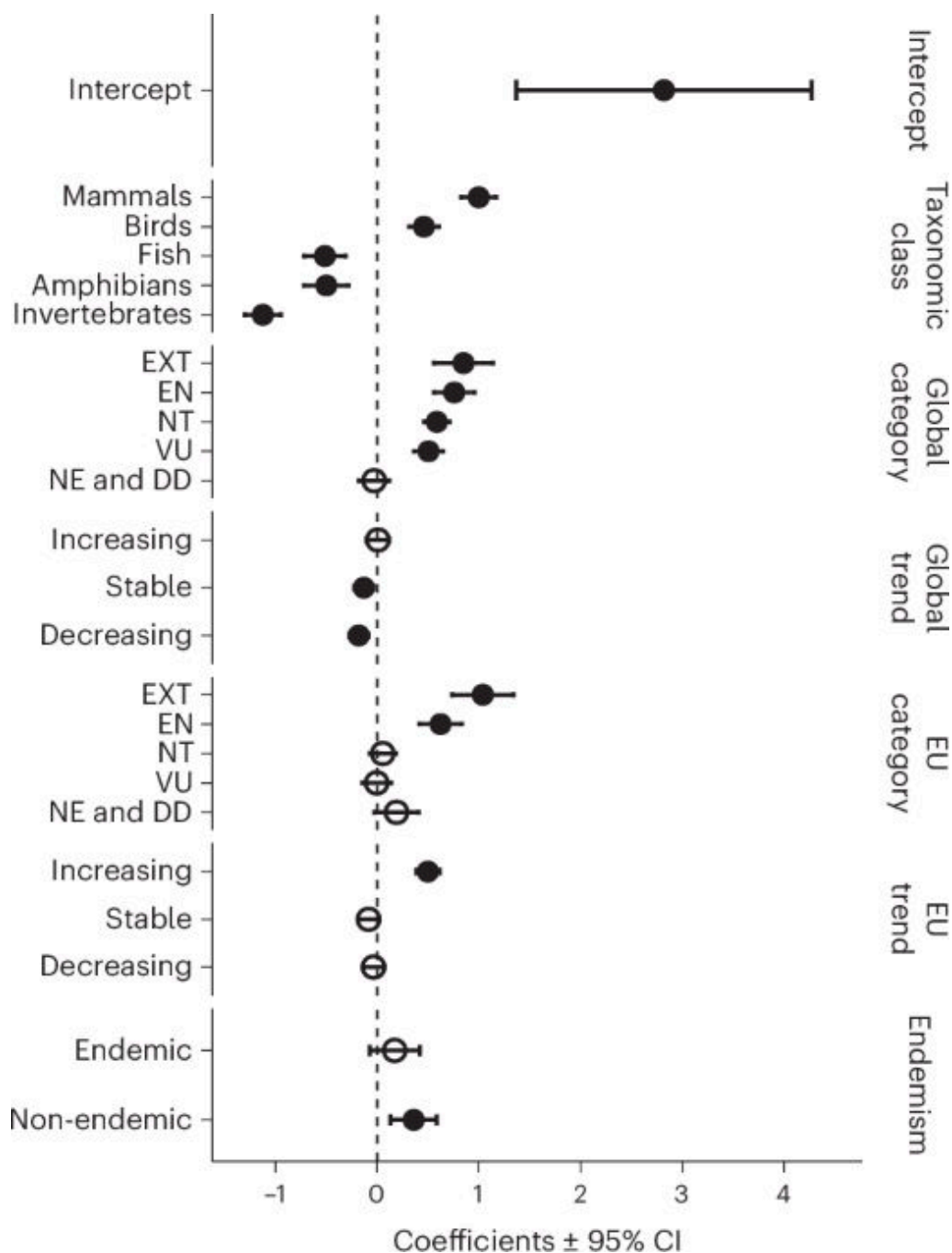


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Ecology & Evolution, Published online: 12 May 2026; doi:10.1038/s41559-026-03066-6 Our analysis of nearly 10,000 legal compensation values across 24 European countries shows that legal price lists for species systematically reflect conservation priorities, taxonomic biases and life-history traits. This analysis reveals hidden criteria that shape regulatory decisions and highlights opportunities to integrate science-based criteria into legal valuation practices.

#### 中文摘要

研究分析欧洲 24 个国家近 10000 条物种法律补偿价值，发现物种法律“价格表”系统性反映保护优先

级、分类群偏差和生活史性状。该分析揭示了影响监管决策的隐性标准，并指出可将科学依据更好纳入法律估值实践。

深度解读

生物多样性法律补偿并非中性技术过程，而会受到社会偏好和制度惯性的影响。若某些分类群被系统性低估，损害赔偿和执法激励也会随之偏移。该研究为建立更透明、更生态学导向的物种价值评估体系提供证据。

DOI: <https://doi.org/10.1038/s41559-026-03066-6>

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环境与气候

27. Synergies between poverty reduction and biodiversity conservation

(减贫与生物多样性保护之间的协同关系)

Nature Sustainability | 2026-05-18 | 【Recent】

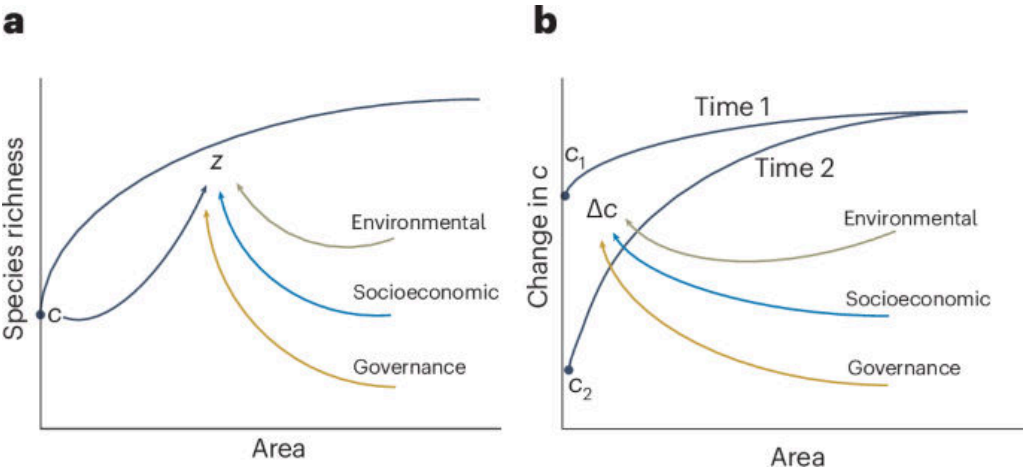


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01830-x Greater household dependence on forest resources is associated with lower tree species diversity in tropical forests, highlighting the interconnected dynamics of poverty and biodiversity in multifunctional landscapes. Forest policies that aim to conserve biodiversity must consider the interconnected dynamics of poverty alleviation and biodiversity conservation to achieve multiple sustainability goals.

中文摘要

研究发现，家庭对森林资源依赖程度越高，热带森林树种多样性往往越低，凸显多功能景观中贫困与生物多样性的相互联系。森林政策若要保护生物多样性，需要同时考虑减贫和保护目标。

深度解读

该研究避免把当地居民生计与保护目标简单对立，而是强调二者之间的结构性联系。高资源依赖可能反映替代生计不足、制度支持不足或市场压力。有效保护需要通过减贫、权益安排和可持续利用降低对森林的高强度依赖。

DOI: <https://doi.org/10.1038/s41893-026-01830-x>

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# 环境与气候

## 28. Advancing biodiversity through poverty solutions (通过贫困解决方案促进生物多样性)

Nature Sustainability | 2026-05-18 | 【Recent】

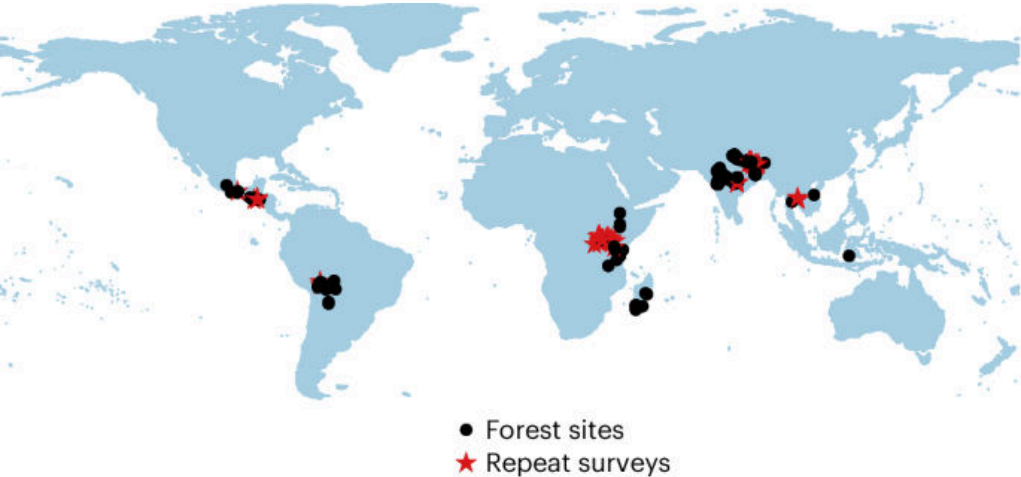


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

### Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01816-9 Forest biodiversity is linked, in many cases, to the myriad ecosystem services that support the livelihoods of human populations. Authors here examine the demographic, socioeconomic and institutional variables that impact forest biodiversity across 322 tropical forests.

### 中文摘要

文章指出，森林生物多样性在许多情况下与支持人类生计的多种生态系统服务相联系。作者考察了322个热带森林中人口、社会经济和制度变量对森林生物多样性的影响。

### 深度解读

这项研究把热带森林保护置于社会—生态系统框架下，而非单纯生态变量解释。人口结构、收入来源和制度安排会改变森林利用强度及其对多样性的影响。它提示保护政策需要与社会治理和生计改善同步设计。

DOI: <https://doi.org/10.1038/s41893-026-01816-9>

# 环境与气候

## 29. Sea-level-driven land conversion amplified by coastal agriculture (沿海农业放大海平面驱动的土地转换)

Nature Sustainability | 2026-05-18 | 【Recent】

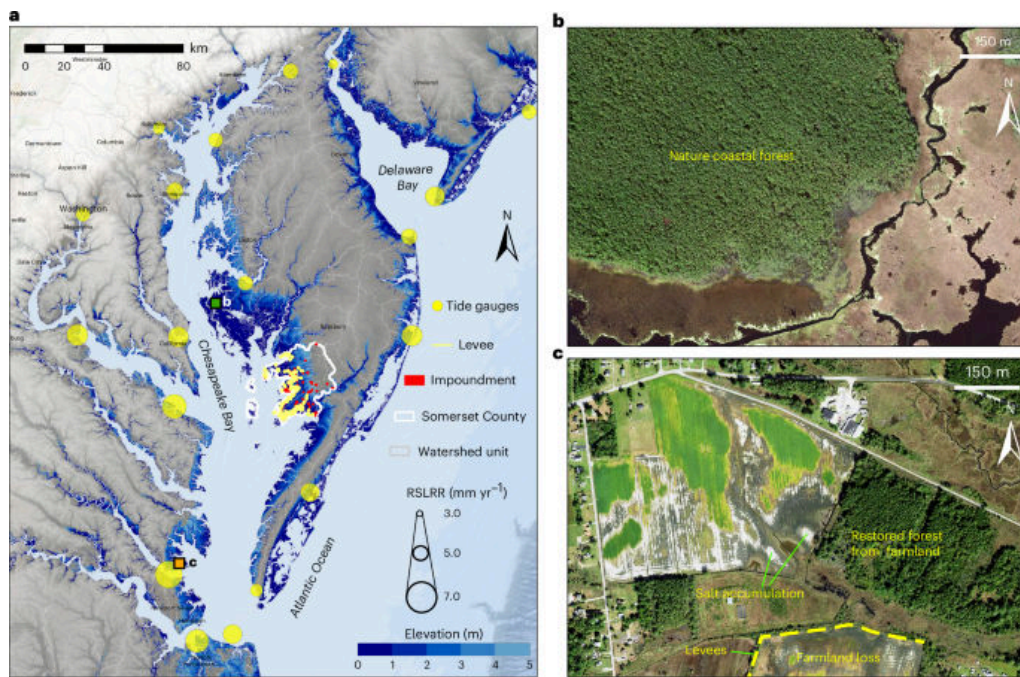


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Sustainability, Published online: 18 May 2026; doi:10.1038/s41893-026-01835-6As sea levels rise, coastal ecosystems and economies are threatened by saltwater intrusion and wetland encroachment. These effects are accelerated by coastal agriculture, highlighting the importance of local management interventions.

## 中文摘要

随着海平面上升，盐水入侵和湿地扩张威胁沿海生态系统与经济活动。研究指出，沿海农业会加速这些影响，凸显地方管理干预的重要性。

## 深度解读

海平面上升的土地影响会被本地土地利用方式显著调节。农业排水、地表改造和淡水管理可能降低系统抵抗盐水入侵与湿地迁移的能力。沿海适应需要把农田管理、湿地缓冲和有序退让结合起来。

DOI: <https://doi.org/10.1038/s41893-026-01835-6>

## 环境与气候

### 30. Emergent constraints on future methane emissions from global wetlands

(全球湿地未来甲烷排放的涌现约束)

Nature Geoscience | 2026-05-19 | 【Recent】



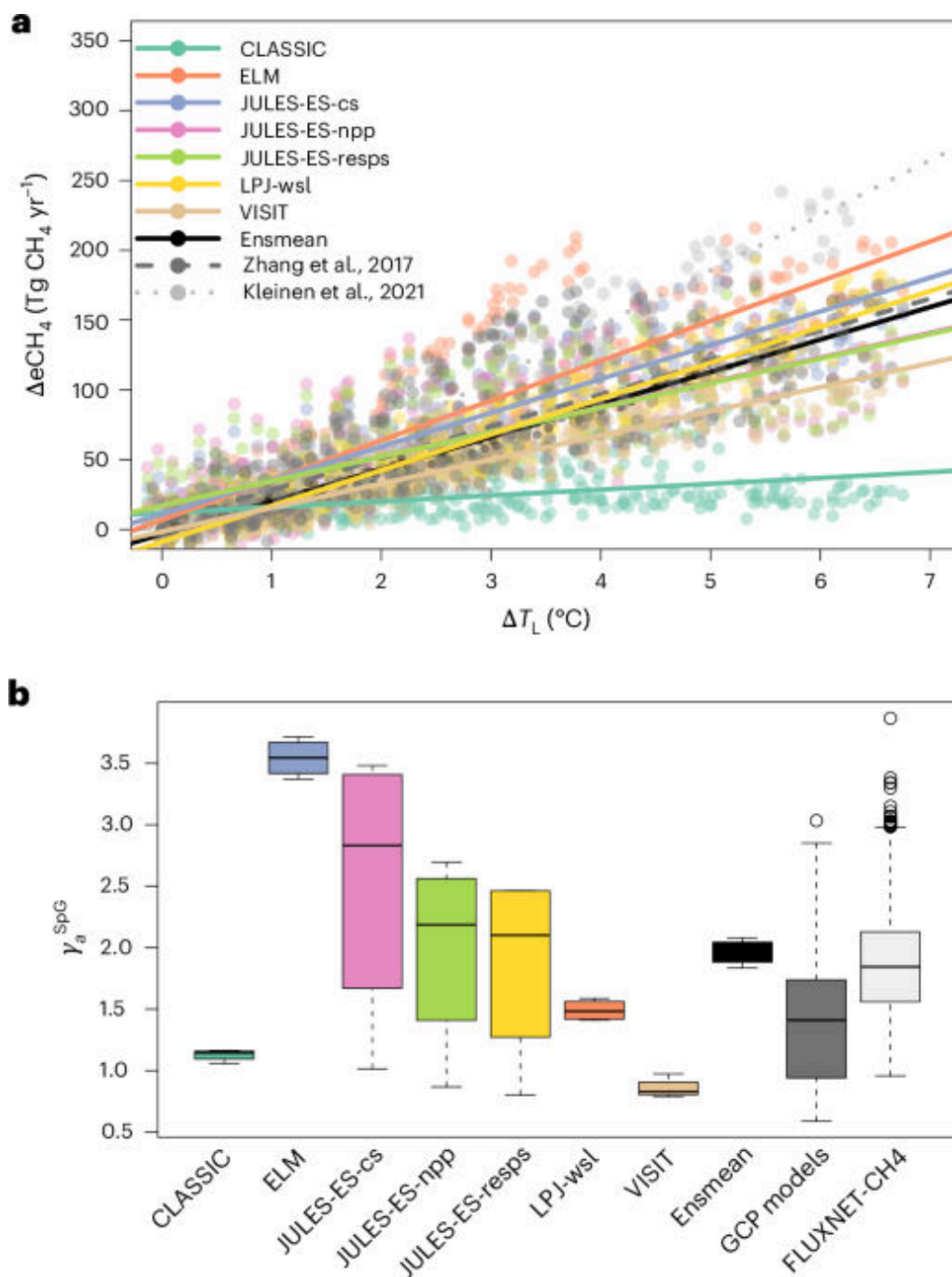


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Geoscience, Published online: 19 May 2026; doi:10.1038/s41561-026-01987-2 Enhanced future methane emissions from global wetlands under warming could substantially offset the emissions reduction goals of the Global Methane Pledge, according to ensemble simulations from terrestrial biosphere models.

#### 中文摘要

基于陆地生物圈模型集合模拟，研究表明变暖下全球湿地未来甲烷排放增强，可能显著抵消《全球甲烷承诺》的减排目标。该文利用涌现约束思路缩小未来排放不确定性。

#### 深度解读

湿地甲烷是气候—碳循环反馈中的关键不确定源。变暖会通过温度、湿地面积、水位和底物供应改变甲烷产生与释放。若自然源增加被低估，人为甲烷减排目标需要更强力度才能实现同等气候效果。

# 海洋与水生

## 31. Chemoproteomics reveals global occurrence of a phytoplankton lyase capable of reconverting progestogens (化学蛋白质组学揭示可再转化孕激素的浮游植物裂解酶在全球存在)

Nature Water | 2026-05-05 | 【Recent】

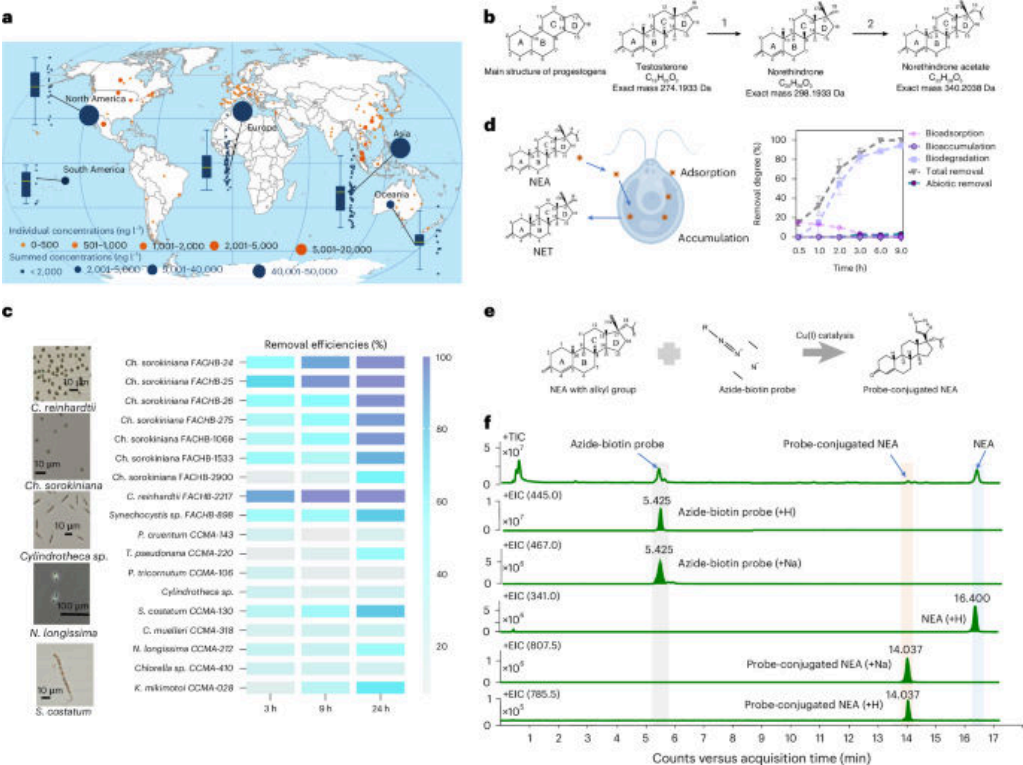


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

### Abstract

Nature Water, Published online: 05 May 2026; doi:10.1038/s44221-026-00646-5 Endocrine-disrupting progesterogens pose risks to ecosystems through their biological activity. A chemoproteomics study shows that many phytoplankton species can transform these chemicals into forms that may be even more biologically active, potentially increasing ecological risk.

### 中文摘要

具有内分泌干扰作用的孕激素会对生态系统产生风险。化学蛋白质组学研究显示，许多浮游植物物种可将这些化学物质转化为可能具有更强生物活性的形式，从而潜在增加生态风险。

### 深度解读

污染物在水生态系统中的风险不仅取决于排放量，也取决于生物转化路径。浮游植物作为初级生产者，可能通过酶促反应改变污染物活性和食物网暴露。该研究提示水质风险评估需纳入生物再活化过程，而非只关注降解。

DOI: <https://doi.org/10.1038/s44221-026-00646-5>

# 植物与农业

## 32. Engineering soil microbiomes for resistance

(工程化土壤微生物组以增强抗扰性)

Nature Food | 2026-05-15 | 【Recent】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

### Abstract

Nature Food, Published online: 15 May 2026; doi:10.1038/s43016-026-01347-8 Soil microbial communities sustain key ecosystem services but are impacted by global change, including warming. Some agricultural soil microbiomes show resistance to perturbation and could be engineered to confer resistance to more fragile systems.

### 中文摘要

土壤微生物群落维持关键生态系统服务，但会受到变暖等全球变化影响。文章指出，一些农业土壤微生物组表现出对抗扰的抵抗力，并可能通过工程化方式把这种抗性赋予更脆弱的系统。

### 深度解读

该文强调微生物组可作为农业和生态适应的主动调控对象。抗扰性可能来自群落冗余、功能互补或长期管理筛选形成的稳定结构。未来应用需要同时评估功能收益、生态安全和跨土壤类型的可迁移性。

DOI: <https://doi.org/10.1038/s43016-026-01347-8>

# 植物与农业

## 33. Agricultural soil microbiomes are structurally and functionally more resistant to warming than adjacent natural ecosystems

(农业土壤微生物组在结构和功能上比邻近自然生态系统更耐受变暖)

Nature Food | 2026-05-15 | 【Recent】

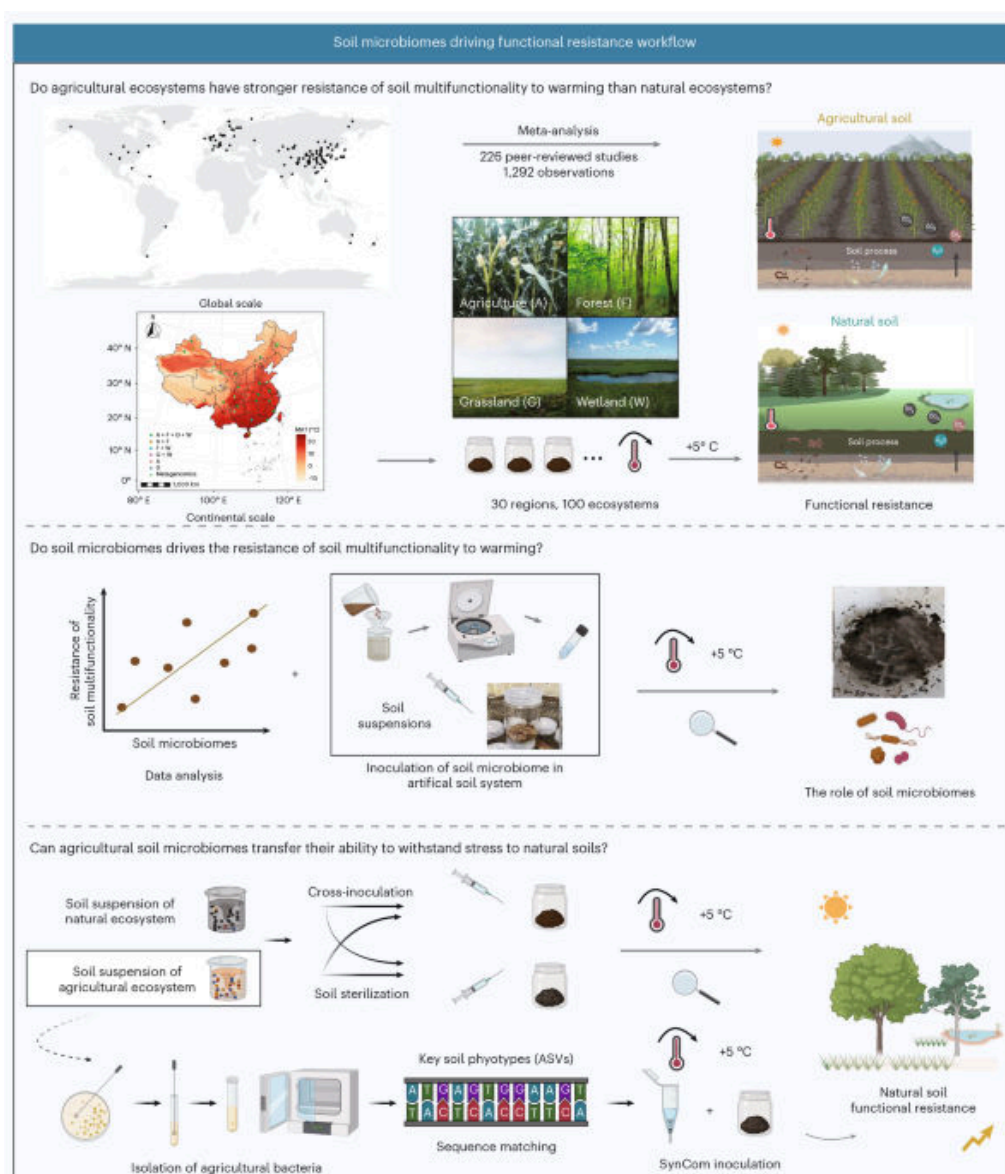


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Food, Published online: 15 May 2026; doi:10.1038/s43016-026-01348-7 Using warming microcosms across 100 paired sites, a global meta-analysis and microbiome manipulation experiments, this study shows that agricultural microbiomes confer higher compositional and multifunctionality resistance to warming and can transfer this resistance to natural soils.

## 中文摘要

研究结合 100 个配对样点的变暖微宇宙实验、全球荟萃分析和微生物组操纵实验，发现农业微生物组对变暖具有更高的组成和多功能性抵抗力，并可将这种抵抗力转移到自然土壤。

## 深度解读

该研究为土壤微生物组抗气候扰动提供了跨尺度证据。农业系统长期承受管理扰动，可能筛选出更耐受温度变化的微生物群落。若可控转移成立，微生物组管理可能成为提升土壤功能稳定性的工具，但需谨慎评估生态外溢效应。

DOI: <https://doi.org/10.1038/s43016-026-01348-7>

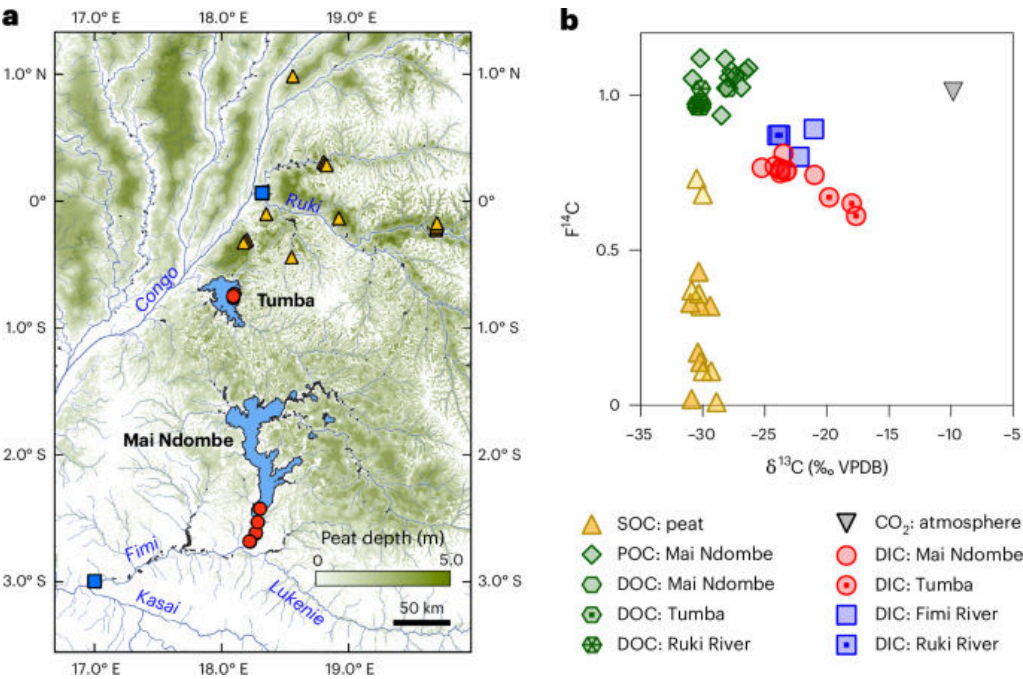


植物与农业

34. Habitat-Mediated Filtering, Rather Than Soil Properties, Shapes Plant Community Diversity on Nelson Island, Maritime Antarctica

(海洋性南极 Nelson 岛植物群落多样性主要受生境过滤而非土壤性质塑造)

Journal of Vegetation Science | Sat, 09 May 2026 05:20:17 -0700 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 3, May/June 2026.

中文摘要

该 Journal of Vegetation Science 论文题名表明，研究关注海洋性南极 Nelson 岛植物群落多样性的驱动机制，认为生境介导的环境过滤比土壤性质更关键。所给摘要仅含期刊卷期信息，具体数据和结论需以原文为准。

深度解读

南极植被处在极端环境边界，微地形、积雪、湿度和暴露条件可能强烈限制物种建立。若生境过滤占主导，气候变暖引起的生境格局变化可能比土壤养分变化更快地重塑群落。该研究对预测极地植被扩张和多样性变化具有参考价值。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70145>

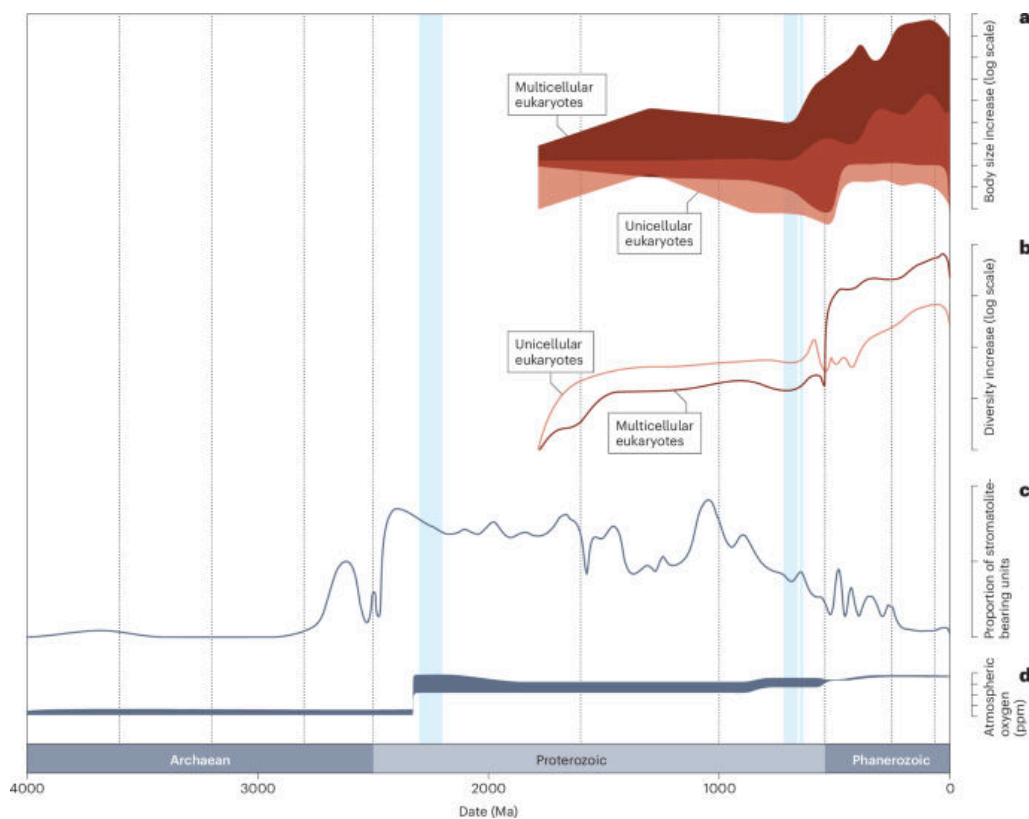
土壤与生物地球化学

35. Benthic phosphorus cycling in the northern Benguela upwelling system: excess P supply and altered pelagic nutrient stoichiometry

(北部本格格拉上升流系统的底栖磷循环：过量磷供应与远洋营养盐化学计量改变)

Biogeosciences | Mon, 11 May 2026 02:09:04 +0200 | 【Recent】





图：Graphical abstract / article figure source:

#### Abstract

Benthic phosphorus cycling in the northern Benguela upwelling system: excess P supply and altered pelagic nutrient stoichiometry Peter Kraal, Kristin A. Ungerhofer, Darci Rush, and Gert-Jan Reichart Biogeosciences, 23, 3253–3277, <https://doi.org/10.5194/bg-23-3253-2026>, 2026 Element cycles in oxygen-depleted areas such as upwelling areas inform future deoxygenation scenarios. The Benguela upwelling system shows strong decoupling of nitrogen and phosphorus cycling due to seasonal shelf anoxia. Anaerobic processes result in pelagic nitrogen loss as  $N_2$ . At the same time, sediments are rich in fish-derived and bacterial phosphorus, with high fluxes of excess phosphate, altering deep-water nitrogen:phosphorus ratios. Such alterations can affect ocean functioning.

#### 中文摘要

研究利用缺氧上升流区理解未来海洋脱氧情景下的元素循环。北部本格拉上升流系统因季节性陆架缺氧而表现出氮、磷循环强烈脱耦：厌氧过程造成远洋氮以  $N_2$  形式损失，同时沉积物富含鱼源和细菌来源磷并释放过量磷酸盐，改变深水氮磷比。

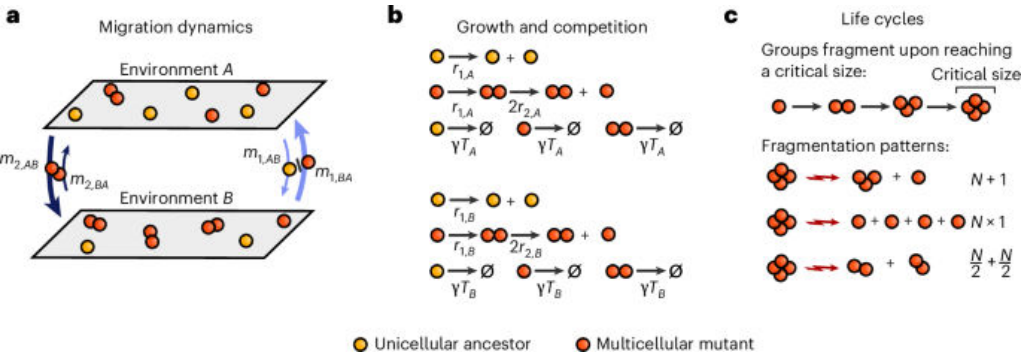
#### 深度解读

该研究揭示缺氧环境中底栖过程可深刻重塑上覆水体营养盐结构。氮损失与磷释放并行会改变初级生产的限制因子，并可能影响浮游生态系统功能。随着全球海洋脱氧加剧，类似机制可能在更多陆架海区增强。

DOI: <https://doi.org/10.5194/bg-23-3253-2026>

## 环境与气候

36. [ASAP] Remote Sensing Enables Basin-Scale Inventories of Coal Mine Methane (遥感支持流域尺度煤矿甲烷清单构建)



图： Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c14976

### 中文摘要

该 Environmental Science & Technology ASAP 论文题名显示，研究关注利用遥感技术开展流域尺度煤矿甲烷排放清单。所给摘要仅提供期刊和 DOI 信息，具体遥感平台、反演方法和排放结果需以原文为准。

### 深度解读

煤矿甲烷是能源系统中重要且可减排的温室气体来源。遥感能够弥补自报清单在空间覆盖和高排放点识别上的不足。流域尺度清单有助于把排放监测从设施级发现推进到区域治理和政策核查。

DOI: <https://doi.org/10.1021/acs.est.5c14976>

## 植物与农业

37. Agri-environmental policies have reduced cropland degradation globally

(农业环境政策已在全球范围降低耕地退化)

Nature Food | 2026-05-18 | 【Recent】

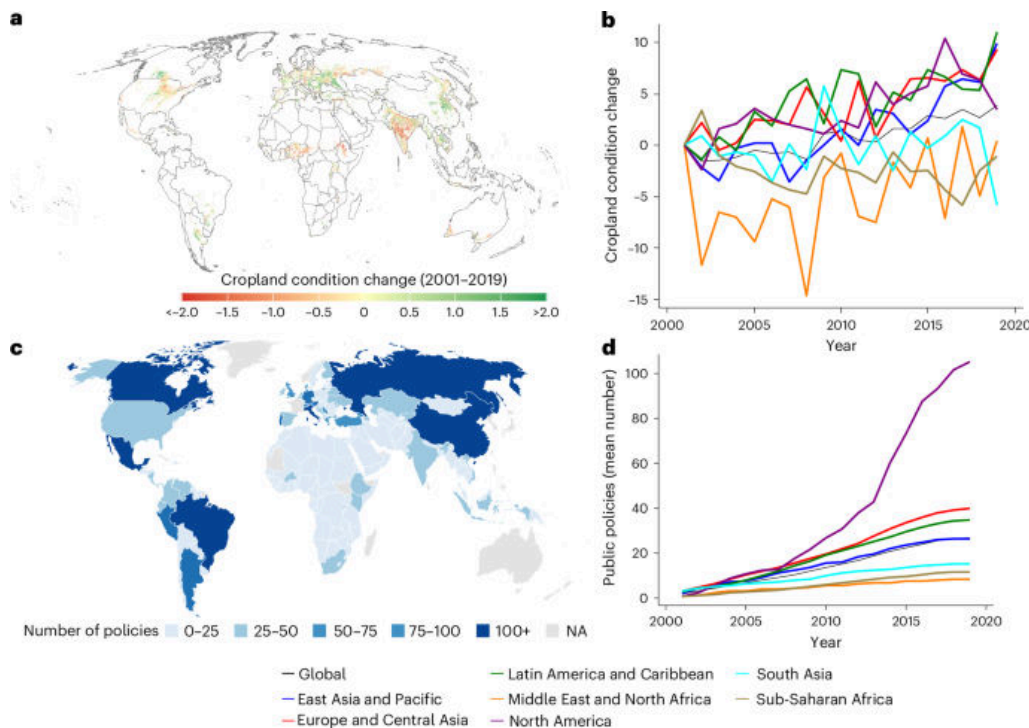


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Food, Published online: 18 May 2026; doi:10.1038/s43016-026-01359-4 Evidence on the effectiveness of national policies against cropland degradation is limited. Using global satellite data from 2001 to 2019, this study addresses this gap and reveals which country characteristics are associated with distinct policy success levels.

## 中文摘要

关于国家政策抑制耕地退化有效性的证据仍有限。该研究利用 2001—2019 年全球卫星数据评估政策效果，并识别与不同政策成功水平相关的国家特征。

## 深度解读

该研究为农业环境政策提供了全球尺度的遥感证据。耕地退化受气候、管理、市场和治理能力共同影响，政策效果也会因国家条件而异。结果可帮助从“是否有政策”转向评估“什么条件下政策有效”。

DOI: <https://doi.org/10.1038/s43016-026-01359-4>

## 保护与生物多样性

38. Climate envelope models (CEMs) identify key factors determining the future distribution of European vertebrates in a changing climate

(气候包络模型识别变化气候下欧洲脊椎动物未来分布的关键因素)

Biodiversity and Conservation | 2026-05-14 | 【Recent】

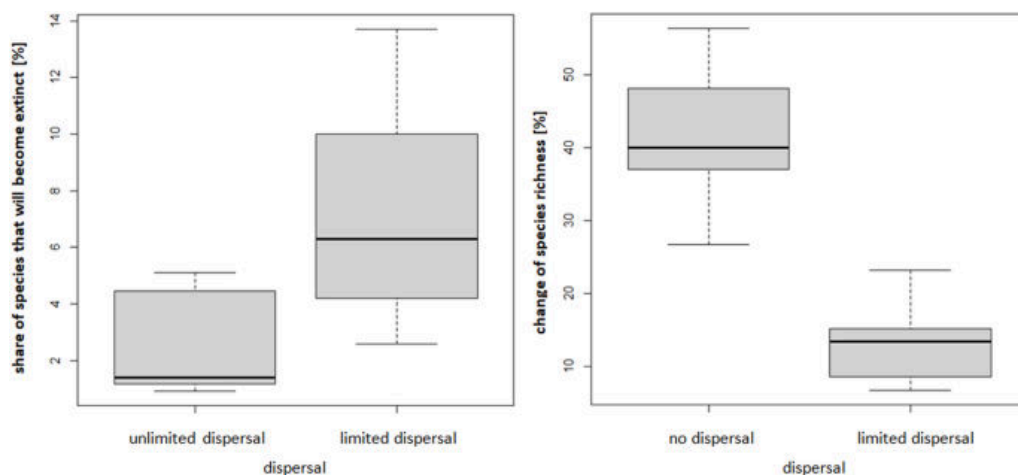


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1007%2F>

## Abstract

Current climate change exerts significant impacts on global ecosystems and is expected to reshape species distributions in the future. Climate Envelope Models (CEMs) are widely used to simulate potential shifts in species ranges under varying climatic conditions. This study aims to quantify the relative importance of key assumptions embedded within CEMs on projections of distributional changes for European vertebrates. This allows us to identify the factors most crucial to the survival of populations and thus to conservation strategies. We analysed variance on published CEM results to estimate effect sizes associated with different modelling assumptions, including the climate model used and the incorporation of dispersal constraints. The capacity for species dispersal emerged as the most i

## 中文摘要

研究评估气候包络模型中关键假设对欧洲脊椎动物未来分布预测的相对重要性。作者通过分析已发表模型结果的方差，比较气候模型选择、扩散限制纳入等假设的效应，发现物种扩散能力是影响预测和保护策略的关键因素之一。

## 深度解读

物种分布预测的不确定性不仅来自气候情景，也来自模型对生物过程的简化。扩散限制决定物种能否追踪未来适宜气候，是保护廊道和辅助迁移决策的核心参数。该研究提醒保护规划应显式处理模型假设，而不是只采用单一预测图。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03373-1>

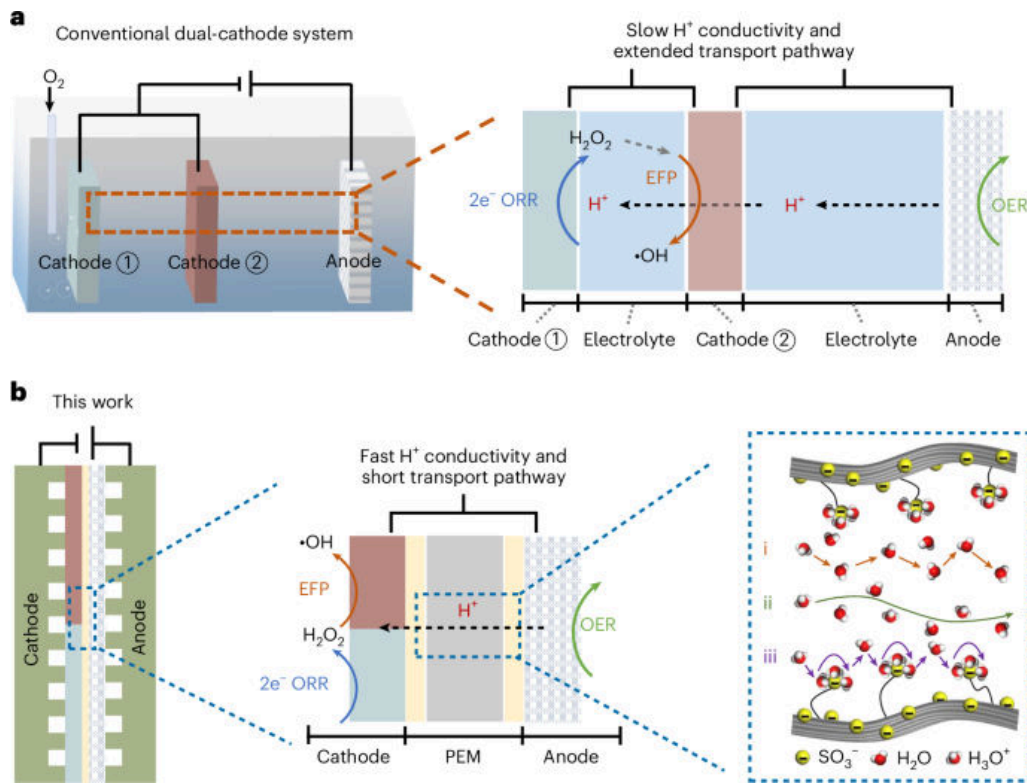
## 遥感与模型

39. Reducing Spatial Bias at Heterogeneous Sites: A Novel Eddy Covariance Filtering Approach Integrating Footprint Modeling and Remote Sensing

(降低异质性站点空间偏差：融合通量足迹模型与遥感的新型涡度协方差筛选方法)

Journal of Geophysical Research - Biogeosciences | Mon, 18 May 2026 22:14:08 -0700 | 【Recent】





图：Graphical abstract / article figure source:

#### Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 5, May 2026.

#### 中文摘要

该 Journal of Geophysical Research: Biogeosciences 论文题名表明，研究提出一种结合通量足迹建模和遥感的新型涡度协方差数据筛选方法，以降低异质性站点的空间代表性偏差。所给摘要仅含期刊卷期信息，具体算法和验证结果需以原文为准。

#### 深度解读

涡度协方差观测常被用于估算生态系统碳水通量，但在土地覆盖异质区域容易受足迹变化影响。将遥感空间信息与足迹模型结合，可更清楚地识别通量观测实际代表的地表单元。该方法有助于提高站点观测与区域尺度模型或卫星产品比较的可靠性。

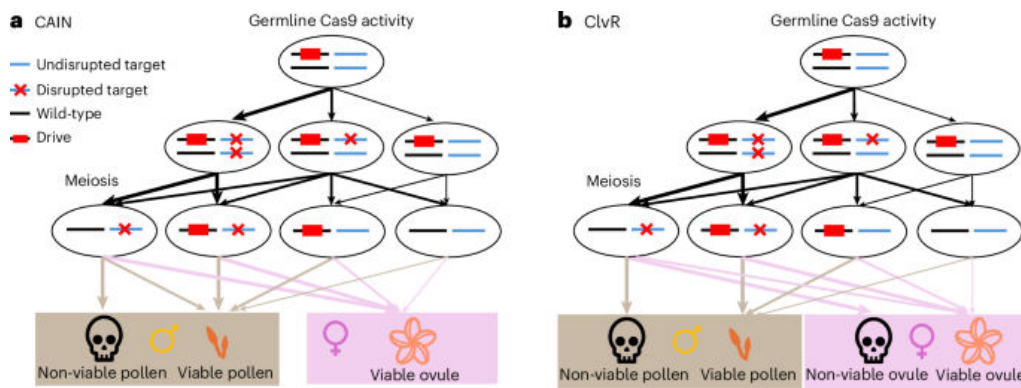
DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009619>

## 生态学核心

### 40. Indirect Interactions Driven by Soil Effects Enable Coexistence Among Competing Plant Species

(土壤效应驱动的间接相互作用促进竞争植物物种共存)

Ecology Letters | Sun, 17 May 2026 05:59:28 -0700 | 【Recent】



图：Graphical abstract / article figure source:

Abstract

中文摘要

该 Ecology Letters 论文题名表明，研究关注土壤效应引发的间接相互作用如何使竞争植物物种实现共存。由于所给摘要为空，具体实验系统、土壤反馈类型和机制强度需以原文为准。

深度解读

植物共存不仅由地上竞争决定，也受植物—土壤反馈和微生物介导过程影响。某一物种改变土壤后，可能间接影响其他竞争者的表现，从而形成稳定化机制。该主题有助于解释群落多样性维持和植被恢复中的物种组合效应。

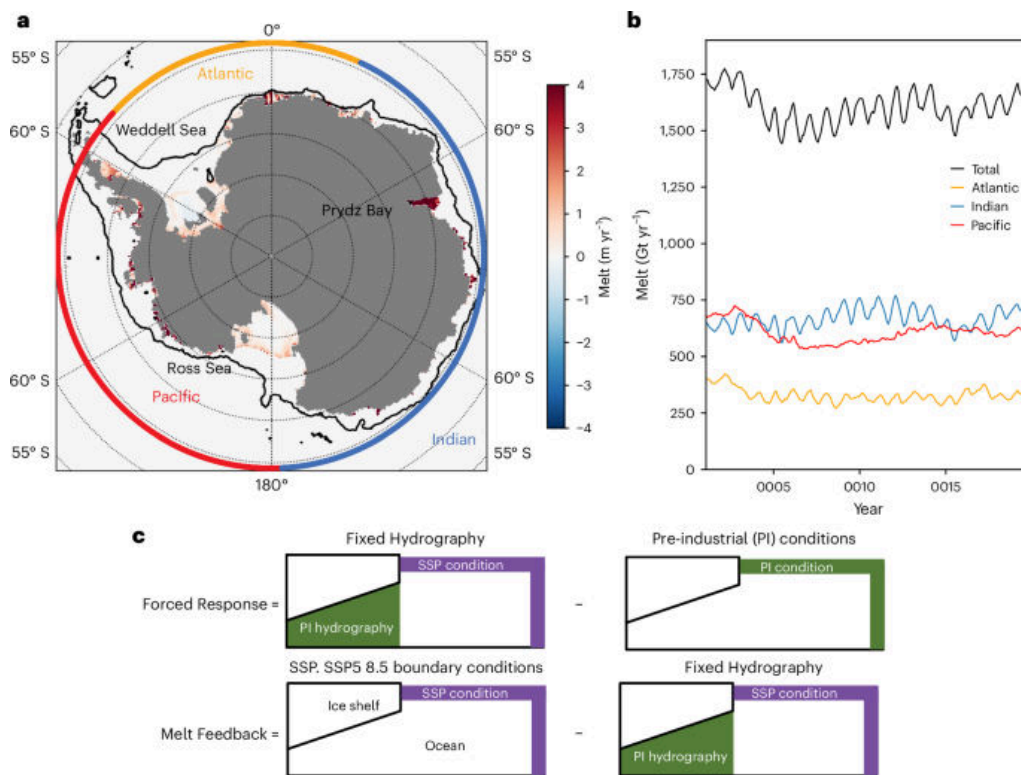
DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70396>

## 环境与气候

41. Antarctic ice-shelf basal melt shaped by competing feedbacks

(竞争性反馈塑造南极冰架底部融化)

Nature Geoscience | 2026-05-15 | 【Recent】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

## Abstract

Nature Geoscience, Published online: 15 May 2026; doi:10.1038/s41561-026-01975-6 Freshwater fluxes from meltwater can intensify Antarctic ice-shelf loss by enabling deep-water intrusions into shelf cavities, while surface freshening may instead form protective layers that limit basal melt, according to future model simulations.

## 中文摘要

未来模型模拟显示，融水淡水通量可通过促进深层水进入冰架腔体而加剧南极冰架损失；但表层淡化也可能形成保护性水层，限制底部融化。冰架基底融化因此受相互竞争的反馈机制控制。

## 深度解读

冰架稳定性取决于海洋分层、热量输送和淡水输入之间的耦合。不同反馈方向相反，意味着局地海洋条件会显著影响冰架对变暖的响应。准确模拟这些机制对预测南极冰盖质量损失和全球海平面上升至关重要。

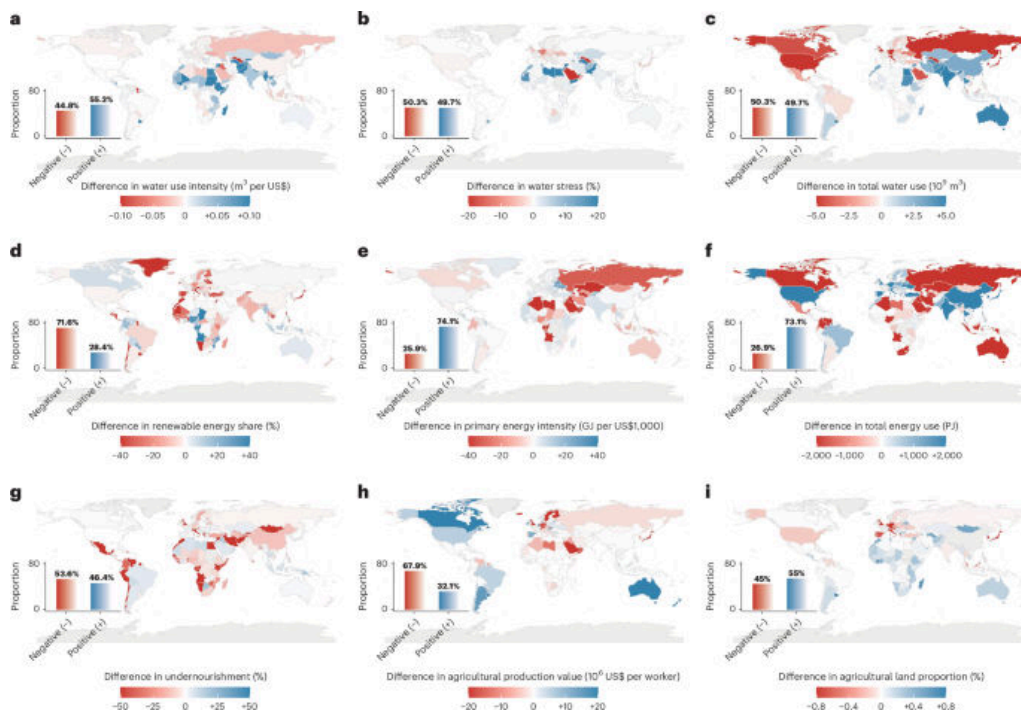
DOI: <https://doi.org/10.1038/s41561-026-01975-6>

## 环境与气候

42. Variations in the root-soil system influence the grapevine holobiont by shaping plant physiology and root microbiome

(根—土系统变异通过塑造植物生理和根际微生物组影响葡萄藤整体生物体)

Science of the Total Environment || 【Recent】



图：Graphical abstract / article figure source:

#### Abstract

Publication date: 25 June 2026 Source: Science of The Total Environment, Volume 1037 Author(s): Massimo Guazzini, Ramona Marasco, Slobodanka Radović, Elisa Pellegrini, Marco Vuerich, Arianna Lodovici, Giorgia Dubsky De Wittenau, Eleonora Paparelli, Gabriele Magris, Laura Zanin, Marco Contin, Elisa De Luca, Daniele Daffonchio, Gabriele Di Gaspero, Fabio Marroni

#### 中文摘要

该 Science of the Total Environment 文章题名显示，研究关注根—土系统差异如何通过影响植物生理状态和根部微生物组，进而改变葡萄藤“整体生物体”（holobiont）。所给摘要主要为出版信息和作者列表，具体实验结果需以原文为准。

#### 深度解读

葡萄藤表现是植物、土壤和微生物共同作用的结果，而非单一品种或施肥因素决定。根际环境会调节水分养分吸收、胁迫响应和微生物互作，从而影响作物健康与品质。该研究对精准葡萄栽培和土壤微生物管理具有应用意义。

DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726005383>

## 海洋与水生

### 43. Urea runoff fuels eutrophication of shallow surface waters in agricultural regions

(尿素径流推动农业区浅层地表水富营养化)

Nature Water | 2026-05-06 | 【Recent】

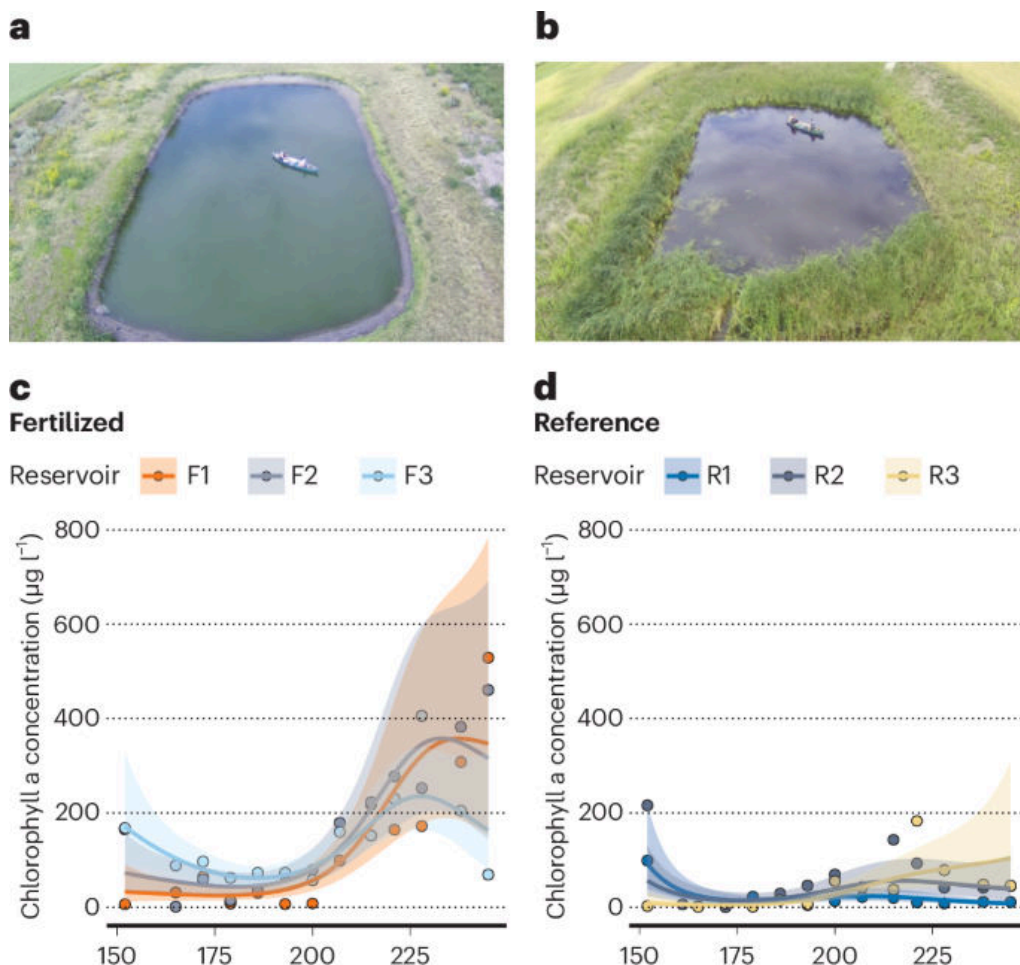


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

#### Abstract

Nature Water, Published online: 06 May 2026; doi:10.1038/s44221-026-00641-w Urea is the world's most widely used nitrogen fertilizer. Elevated urea use can induce detrimental eutrophication of shallow surface waters in agricultural regions, accelerating water quality deterioration globally in these regions.

#### 中文摘要

尿素是全球使用最广泛的氮肥。研究指出，在农业区，尿素使用升高会诱发浅层地表水有害富营养化，并加速这些地区水质恶化。

#### 深度解读

传统富营养化管理常强调硝态氮和铵态氮，但尿素作为有机氮形态同样可能直接刺激藻类和微生物过程。浅水系统水动力弱、光照充足，对外源氮输入尤其敏感。该研究支持在农业面源污染控制中单独关注尿素径流和施肥时序。

DOI: <https://doi.org/10.1038/s44221-026-00641-w>